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Physics-Informed Ensemble Learning for city-center grid cell temperature prediction during thermal extremes☆

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ABSTRACT

Accurate prediction of city-region temperature is crucial for urban sustainability, which impacts energy management and public health. Current models struggle with spatio-temporal non-linearities and exhibit poor performance during extreme events due to data scarcity. We propose Physics-Informed Ensemble Learning (PIEL-NET). This novel hierarchical framework integrates a physics-terrain fusion agent with a regime-specialized agent trained under Regime Adaptive Focal Loss (RAFL). This architecture enables dynamic focus on underrepresented extreme cases where physical models underperform, distinguishing it from conventional physics-informed or ensemble methods. Our model processes the urban core and eight surrounding grid cells at 100 km resolution to capture advection context. Evaluated on decade-long hourly data from four global cities, PIEL-NET achieves a 69.2% error reduction below −7.4 °C in extreme cold conditions and attains a 0.89 Dice score for extreme event warnings, significantly outperforming five state-of-the-art benchmarks.

1. Introduction

Human interventions drive frequent and intense temperature extremes, which pose significant challenges to urban environments. These fluctuations highlight the need for short-term temperature prediction (TP), particularly in the context of sustainable city planning and management. Rising temperatures cause non-linear increases in electricity demand (Yang et al., 2024), especially in urban areas, while also affecting health (Chen et al., 2025), agriculture (Hogan and Schlenker, 2024), food security (Narayan et al., 2024) and water resources (Ray and Tikuye, 2025). Accurate TP is essential to manage energy consumption and ensure grid stability, particularly in smart cities and resilient environments. Such predictions support energy efficiency, climate change mitigation, and sustainable urban development. Timely TP also enables improved decision-making for managing infrastructure, air quality and urban health, enhancing resilience against the evolving challenges of climate change.

Existing deep learning models are primarily data-driven and have several limitations, as they can be physically inaccurate or may violate physical principles when making predictions. Furthermore, these models are usually trained using mean absolute error (MAE) and mean squared error (MSE) objectives, and they learn from datasets that inherently reflect diurnal patterns, where temperatures rise with sunlight and decline after sunset. Moreover, each geographical location exhibits distinct temporal characteristics influenced

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by Earth's rotation and seasonal cycles, for instance May and June correspond to winter months in the Southern Hemisphere while representing summer in the Northern Hemisphere. Consequently, data-driven models primarily learn these recurring seasonal and diurnal patterns. However, climate change introduces increasing thermal extremes that disrupt these established patterns, and most existing models struggle to predict such extreme events because their training objectives prioritize learning ordinary daily patterns while treating extremes as statistical outliers. These limitations highlight the need for approaches that improve the focus on rare high-impact thermal extremes while maintaining prediction accuracy under normal conditions.

To address these limitations, we introduce PIEL-NET, a physics-informed hierarchical ensemble model designed to integrate physical principles with data-driven learning for robust temperature prediction. We first develop a Physics-Terrain Fusion Agent to establish a physically-consistent foundation for the model. This agent uses a simplified advection–diffusion equation to generate a baseline forecast that adheres to atmospheric dynamics and accounts for local terrain effects. To capture complex spatio-temporal patterns, we design a Spatio-Temporal Convolutional LSTM (STCL) agent. We enhance this core network with residual skip connections to mitigate vanishing gradients and enable stable training in deep layers. This STCL agent is trained in two strategic stages. It first learns general meteorological patterns using a standard MAE objective. It then undergoes specialized refinement using a novel Regime-Adaptive Focal Loss (RAFL) that forces the model to concentrate on correcting errors during rare thermal extremes. Finally, a hierarchical ensemble framework fuses the outputs from the physics agent, the baseline STCL model, and the regime-specialized expert. This architecture delivers a unified solution that maintains high accuracy for common seasonal and diurnal patterns while achieving superior performance during disruptive extreme events. The primary contributions are as follows:

- **Training regime definition**: Compute residuals from the Physics-Terrain Fusion Agent using composite RMSE-MAE error and apply fuzzy membership to low, medium, and high error regimes, producing continuous sample weights that focus learning where the physics prior underperforms.
- **Regime-Adaptive Focal Loss (RAFL)**: Incorporate regime membership through adaptive weighting with stability terms and focal components, retraining the STCL baseline specifically on hard examples where previous agents fail during extreme temperatures.
- **Hierarchical adaptive fusion** The framework first combines wind-driven heat transfer modeling with terrain-aware feature interaction through the Physics-Terrain Fusion Agent, then fuses this physics-constrained foundation with the Spatio-Temporal Convolutional LSTM baseline predictions and regime-specialized expert outputs using ensemble deep Random Vector Functional Link with skip connections, median aggregation, and closed-form output weights to dynamically balance complementary strategies during inference.

The rest of the paper is organized as follows, with Section 2 reviewing existing work and Section 3 presenting the proposed methodology, while Section 4 reports results and comparisons. Section 5 discusses overall findings and Section 6 concludes the study.

2. Literature review

Temperature prediction (TP) methods comprise four groups that include physical methods, statistical methods, learning-based methods that include machine learning and deep learning, and hybrid models. Physical methods, including numerical weather prediction and urban-canopy models, simulate atmospheric physics to forecast temperature (Skamarock et al., 2019; Kusaka et al., 2001). However, their performance is constrained by sensitivity to initial conditions, parameterizations, and high computational cost, which creates a need for more adaptable and efficient forecasting techniques (Zhang et al., 2013; Lee et al., 2011). In response to these constraints, statistical models reduce computational burden and relax strong physical assumptions while retaining operational clarity. Statistical models, including Autoregressive Moving Average (ARMA) (Hasan et al., 2023), Autoregressive Integrated Moving Average (ARIMA) (Chen et al., 2024) and Kalman Filter (Feifei et al., 2024) methods, provide simplicity and interpretable results. However, these approaches assume linearity and stationarity, which often do not hold for temperature series with nonlinear trends or seasonality, and ARIMA requires careful manual parameter tuning that takes time and risks error.

Classical machine-learning baselines, Support Vector Regression (SVR) and Random Forest (RF), address time-series forecasting, including temperature prediction. SVR learns a large-margin regressor in a kernel space and fits smooth responses when features are well scaled and the radial-basis kernel is tuned (Chevalier et al., 2011). It often outperforms linear baselines on medium-sized datasets with smooth nonlinear structure. However, mis-tuned hyperparameters and kernel width degrade accuracy, and training cost rises with sample size. RF aggregates bootstrapped trees and mitigates these issues. It handles mixed-scale predictors with minimal preprocessing and often surpasses SVR under noisy interactions and heterogeneous covariates (Suthar et al., 2024). Yet RF produces piecewise-constant outputs, cannot extrapolate beyond the training range, and treats lagged windows as independent, which weakens long memory. Both SVR and RF minimize average error and often under-detect rare extremes even when overall performance improves (Chen et al., 2022).

An Artificial Neural Network (ANN) overcomes these limits by learning hierarchical features end-to-end. It captures higher-order interactions that RF and SVR only approximate (Astsatryan et al., 2021). With larger datasets and multimodal input, an ANN models couples among humidity, wind, and pressure more effectively. Paired with a Recurrent Neural Network (RNN) or Long Short-Term Memory (LSTM) unit, it exploits temporal context that tree and margin methods ignore. This capacity explains why ANN often outperforms RF and SVR on average error in prediction tasks (Suthar et al., 2024). Tradeoffs remain because training stays sensitive to initialization and regularization. These models allocate capacity to frequent regimes and underweight rare extremes. These limits motivate regime-aware weighting and a light physics prior to guide learning where data are sparse.

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) (D'Orazio et al., 2023) networks are widely used deep learning models for time series prediction due to their ability to capture temporal dependencies. RNNs, however, face challenges such as the vanishing gradient problem when modeling long sequences, which limits their capacity to capture long-term dependencies effectively. LSTMs (Hochreiter and Schmidhuber, 1997), designed to address this issue, perform better in modeling such dependencies by using gating mechanisms that help retain important information over extended periods. In TP, these models are particularly effective in identifying complex temporal trends and seasonal patterns. In Yang et al. (2022), the authors introduce a framework for urban temperature prediction (UTP) that integrates high-spatiotemporal Internet of Things data with weather station observations, utilizing LSTM. While including variables such as temperature, humidity, wind speed and ultraviolet index improves prediction accuracy, the addition of the ultraviolet index and cloud cover raises the overall hardware costs, particularly when deployed on edge devices.

More recently, transformer architectures draw interest for time-series prediction (Hu et al., 2025; Zhu et al., 2024). These models use self-attention, which captures long-range dependencies and global structure. Transformers are commonly categorized by their tokenization scheme. In point-wise tokenization, each token is a time step that bundles all variates, preserving instantaneous cross-variable correlation while attention links distant times (Vaswani et al., 2017). In patch-wise tokenization, each token is a short contiguous segment of a single variate, which compresses local temporal structure and shortens the effective sequence, improving efficiency but risking weaker instantaneous cross-variable coupling unless later layers fuse patches (Nie et al., 2022). In variate-wise tokenization, each token is the entire history of one variate, which emphasizes long temporal dependencies within that series while attention exchanges information across variates (Liu et al., 2023). However, like other deep-learning models, transformers are data-driven and often underperform on thermal extremes because rare events are scarce in training sets (Camps-Valls et al., 2025). PIEL-NET improves on transformer baselines in two ways, the light physics prior supplies advective context that data-only models need many parameters to learn, and the regime adaptive loss increases gradient weight on rare extremes so tails remain visible when MAE or MSE would underweight them. The hierarchical fusion then lets the data pathway dominate when the physics prior underperforms, which maintains city-center grid cell forecast performance while avoiding the heavy memory footprint of full self attention. In addition, the Spatio Temporal Convolutional LSTM (STCL) agent stacks ConvLSTM layers in a cascaded design, which captures cross variable nonlinear interactions and local spatio-temporal couplings, whereas attention models often emphasize broad holistic patterns, the cascade learns interactions across scales without the parameter overhead of deep self attention.

Hybrid models (Kumar et al., 2019; Wang et al., 2024; Saber et al., 2025; Hou et al., 2022) combine the strengths of different methods to improve accuracy and robustness. Kumar et al. (2019) present a hybrid model that combines Particle Swarm Optimization (PSO) and Extreme Learning Machine (ELM) for short-term TP using ambient sensors. The model optimizes ELM parameters with PSO to enhance prediction accuracy. Similar terrain-dependent hybrid model optimization approaches are proposed for other environmental variables, such as wind speed (Aslam et al., 2025a,d,b). Uluocak and Bilgili (2024) propose using LSTM for feature extraction and CNN for feature refinement before predicting daily temperature. However, they optimize the model's hyperparameters using a trial-and-error method, which makes the model terrain-dependent and adds an extra layer of hyperparameter optimization. These approaches require optimizing the model's hyperparameters individually for each location before training, demanding significant computational resources and reducing feasibility. A more common method involves using CNN for feature extraction and LSTM for prediction, as shown in Farhangmehr et al. (2025) for soil TP. Nevertheless, hybrid deep models such as CNN LSTM and ConvLSTM face two issues, depth causes gradient attenuation that blunts abrupt transitions, and average error losses without physics bias learning toward frequent regimes. This study adds a 3D convolutional front end and residual skip connections across ConvLSTM blocks to stabilize training and retain high-frequency temporal content. It also introduces a light advection and terrain prior with a two-stage objective, MAE followed by RAFL on physics-informed residuals, which redirects the gradient to tail events while keeping accuracy on typical hours.

Physics-informed and hybrid models often degrade when the governing equations are simplified or when training data are sparse on coarse grids (Aslam et al., 2025c, 2024). In convection-dominated and multi-scale regimes and near sharp gradients such as boundary layers, PINNs stall or converge to inaccurate solutions, the PDE residual hardens the loss landscape, and errors remain large even on modest problems (Krishnapriyan et al., 2021). Neural tangent kernel analysis shows different convergence rates for the data misfit and the residual physics, weight choice steers the optimization path, overweighting the physics term reduces data fidelity and degrades accuracy, adaptive reweighting alleviates, but does not remove this sensitivity (Wang et al., 2022). When the residual uses integral losses, naive Monte Carlo or deterministic approximations introduce bias that grows with sparse sampling, the learned solution then shifts away from the target even when the network class is expressive (Saleh et al., 2024). In hybrid corrector schemes, the learned correction depends on the base physics model, which shows noticeable errors outside its calibration domain, the composite can extrapolate poorly when the prior is misspecified (Bock et al., 2021). These limits match the setting of temperature extreme, they motivate a design that treats the physics path as a prior, learns residuals with regime focus, and fuses experts so the data pathway dominates when the physics prior underperforms. Together, these developments show that existing methods optimize mean behavior and often underdetect extremes in many sites, which motivates a hierarchical fusion that couples a physics prior with regime-aware learning in the subsequent method section.

3. Proposed model

This work proposes a PIEL-NET for temperature prediction. The input tensor $X \in \mathbb{R}^{B\times T\times F\times \Theta}$ is built from hourly spatio-temporal records collected at the city center and at peripheral sites located d kilometers from the center along eight directions $\{\mathrm{N}, \mathrm{NE}, \mathrm{E}, \mathrm{SE}, \mathrm{S}, \mathrm{SW}, \mathrm{W}, \mathrm{NW}\}$, with C denoting the center. At each time step the core meteorological features are temperature,

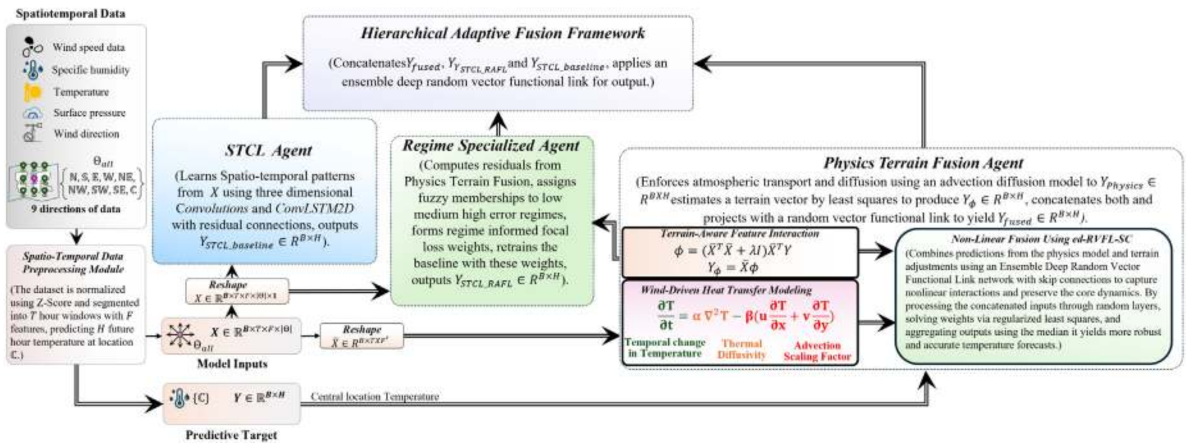

Fig. 1. Architecture of PIEL-NET showing component integration.

humidity, wind speed, wind direction, and air pressure. This work appends four calendar features (year, month, day, hour) as auxiliary inputs when used. Features occupy the F axis and locations occupy the Θ axis, with $\Theta = 9$ when the center is included. The target is to predict $\hat{Y} \in \mathbb{R}^{B\times H}$ using the input X, where H is the prediction horizon.

The pipeline has three agents and a fusion module. Physics Terrain Fusion Agent produces $Y_{\text{physics}} \in \mathbb{R}^{B\times H}$ using an advection–diffusion model and produces $Y_{\phi} \in \mathbb{R}^{B\times H}$ using a terrain vector estimated by least squares, then concatenates these outputs and passes them to a random vector functional link to yield $Y_{\text{fused}} \in \mathbb{R}^{B\times H}$. The spatio-temporal convolutional long-short-term memory agent (STCL) is the baseline data model that uses three-dimensional convolutions and convolution LSTM layers with residual connections to produce $Y_{\text{STCL\_baseline}} \in \mathbb{R}^{B\times H}$. The regime specialized module computes residuals from the Physics Terrain Fusion agent, applies fuzzy membership to low, medium, and high error regimes to derive weights for a regime-informed focal loss, retrains the baseline with this loss, and produces $Y_{\text{STCL\_RAFL}} \in \mathbb{R}^{B\times H}$. The fusion module concatenates the three outputs and applies an ensemble deep random vector functional link to produce $\hat{Y} \in \mathbb{R}^{B\times H}$. The block diagram appears in Fig. 1, and subsequent subsections describe each module.

3.1. Physics-Terrain fusion agent

This agent contains three submodules. The wind-driven heat transfer modeling submodule uses an advection–diffusion model to predict temperatures at central locations. Next, the Terrain-Aware Feature Interaction submodule employs a least squares solution to calculate a vector ϕ for temperature prediction. Finally, a fusion module combines predictions from both submodules to generate future temperature forecasts.

3.1.1. Wind-Driven heat transfer modeling

Advection and diffusion are two processes that model the temperature at a given point. Advection, driven by wind, transports temperature from one region to another, changing the temperature at a point as air masses move. This is captured by the term:

$$-\beta\left(u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y}\right), \tag{1}$$

where u and v represent the wind components in the x and y directions, respectively, and β scales the strength of the advection effect. Diffusion, on the other hand, spreads temperature due to molecular motion, causing heat to equalize over time. This is represented by the term:

$$\alpha\nabla^2 T, \tag{2}$$

where α is the thermal diffusivity, and $\nabla^2 T$ is the Laplacian of temperature. Together, these processes combine into the following equation that governs temperature evolution:

$$\frac{\partial T}{\partial t} = \alpha\nabla^2 T - \beta\left(u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y}\right). \tag{3}$$

This equation models how temperature changes over time by accounting for the transport of heat through wind and the natural spreading of heat via diffusion. Given that we have discrete data in both space and time, we have wind speed, wind direction, and temperature values for a center and 8 cardinal directions, each sampled 100 km away, at 1-h intervals over the previous T hours in each training example. Each grid point at time n is denoted as $T^n_{i,j}$, where i and j refer to the spatial indices and n refers to the time step. The temperature is updated iteratively over time for the next H hours. To discretize the equation, this work uses finite differences. The time derivative is approximated as:

$$\frac{\partial T}{\partial t} \approx \frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t}, \tag{4}$$

where Δt is the time step (in this case 1 h). The spatial derivatives in the advection term are approximated using central differences:

$$\frac{\partial T}{\partial x} \approx \frac{T_{i+1,j}^{n} - T_{i-1,j}^{n}}{2D}, \quad \frac{\partial T}{\partial y} \approx \frac{T_{i,j+1}^{n} - T_{i,j-1}^{n}}{2D}. \tag{5}$$

where, D is the spatial distance between adjacent grid points. The diffusion term is approximated using the central difference for the second spatial derivative:

$$\nabla^2 T \approx \frac{T_{i+1,j}^{n} + T_{i-1,j}^{n} + T_{i,j+1}^{n} + T_{i,j-1}^{n} - 4T_{i,j}^{n}}{D^2}. \tag{6}$$

Putting these approximations together, the temperature at the next time step is computed as:

$$T_{i,j}^{n+1} = T_{i,j}^{n} + \Delta t \left[\alpha \frac{T_{i+1,j}^{n} + T_{i-1,j}^{n} + T_{i,j+1}^{n} + T_{i,j-1}^{n} - 4T_{i,j}^{n}}{D^2} - \beta \left(u \frac{T_{i+1,j}^{n} - T_{i-1,j}^{n}}{2D} + v \frac{T_{i,j+1}^{n} - T_{i,j-1}^{n}}{2D} \right) \right], \tag{7}$$

where $T_{i,j}^{n+1}$ is the temperature at grid point (i, j) and time step $n+1$, and $T_{i,j}^{n}$ is the temperature at the previous time step. The terms $T_{i+1,j}^{n}$, $T_{i-1,j}^{n}$, $T_{i,j+1}^{n}$, and $T_{i,j-1}^{n}$ represent the temperatures at neighboring grid points in the east, west, north, and south directions, respectively. The gradients of temperature in the x- and y-directions are used to calculate the advection and diffusion terms.

The advection–diffusion equation models the primary physical mechanisms for short-term surface temperature change at the mesoscale, which are horizontal heat transport by wind and turbulent dispersion. This approach omits processes like radiative transfer and latent heat fluxes to avoid model overparameterization and because their required input data is not consistently available at the necessary resolution across all cities in the dataset. This simplification means the physical model alone performs less accurately during events dominated by vertical processes or localized phenomena. The subsequent data-driven components in the PIEL-NET architecture learn the residuals between this physical prior and observed data, which corrects for these systematic omissions and ensures robust performance even during extreme events.

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Once the parameters are optimized, the model can predict future temperature by iteratively updating the temperature at each grid point for the next H hours. Given the discrete data for wind speed, wind direction, and temperature at the center and 8 cardinal directions (100 km away), the model updates the temperature field based on the advection and diffusion processes, producing predictions for the future temperature. The results shown in this work are based on the test set, where the model is applied to unseen data to evaluate its ability to generalize and predict the temperature for future hours. The model combines physical processes (advection and diffusion) with data-driven optimization to provide accurate and physically consistent predictions for short-term temperature forecasting.

3.1.2. Terrain-Aware feature interaction

Historical features from both the central location and the surrounding terrain are essential for predicting temperature because the temperature at the central location depends not only on its own atmospheric conditions but also on the conditions of neighboring regions. Wind patterns, temperature gradients, air pressure, and humidity from the surrounding areas influence the temperature at the central location. These interactions can be modeled as *terrain-mediated effects*, where the surrounding environment impacts the future temperature at the central location. Therefore, incorporating features from both the center and the surrounding terrain is necessary for accurate temperature prediction.

The input matrix X is reshaped into a new matrix $\bar{X} \in \mathbb{R}^{B \times F'}$, where $F' = T \times F \times \theta$. This reshaping step flattens the features over T time steps and θ surrounding directions into a single vector of features for each example. After reshaping, the model computes ϕ, which represents the set of coefficients that map the historical features to future temperature predictions. This is done by solving the regularized least squares problem:

$$\phi = \left(\bar{X}^T \bar{X} + \lambda I\right)^{-1} \bar{X}^T Y, \tag{8}$$

where $\bar{X}$ is the reshaped matrix of features, Y is the matrix containing the future temperature values at the central location, λ is the regularization parameter that helps prevent overfitting, and I is the identity matrix. Once computed, the matrix ϕ encodes the relationships between the features from both the central location and the surrounding terrain and the future temperature at the central location. It indicates how each feature from the terrain influences the future temperature prediction. After learning ϕ from the training data, it can be applied to new test data to predict future temperatures. The predicted future temperature values are given by:

$$Y_\phi = \bar{X}\phi, \tag{9}$$

where Y_ϕ represents the predicted future temperatures at the central location, and $\bar{X}$ contains the reshaped features from the test data. The learned matrix ϕ is used to compute the predictions by mapping the test features to the future temperature values. Thus, ϕ represents the learned relationships between the historical features and the future temperature at the central location. Once learned from the training set, it can be used to predict future temperatures by applying it to new test data, accounting for the terrain-mediated effects in the prediction process. error weighting.

3.1.3. Nonlinear fusion via ensemble deep random vector functional link with skip connections(ed-RVFL-SC)

The fusion mechanism integrates the predictions from the physics-based model Y_{physics} and the terrain-aware corrections Y_ϕ through ed-RVFL-SC network. The physics-based model's error tends to grow exponentially as the prediction horizon increases, limiting its ability to provide accurate long-term forecasts. However, it has been observed that incorporating physics-based predictions can help improve the performance of the terrain-aware feature modeling module, particularly in cases like the ablation study on the Tucman dataset, where the fusion of both models outperforms either individual model. By combining the outputs from both the physics and terrain models, we obtain better predictions than relying on them separately. In this approach, the concatenated input $Z = [y_{\text{physics}}, Y_\phi] \in \mathbb{R}^{B\times 2H}$ is processed through L hierarchical layers. Each layer's output is computed as follows:

$$H^{(i)} = \begin{cases} g(ZW_i + b_i) & i = 1 \\ g(ZH^{(i-1)}W_i + b_i) & 2 \le i \le 3 \\ g([Z; H^{(i-3)}]H^{(i-1)}W_i + b_i) & i > 3 \end{cases} \tag{10}$$

where $W_i \in \mathbb{R}^{d_{\text{in}}\times d}$ and $b_i \in \mathbb{R}^d$ are the randomly initialized weights and biases, $g(\cdot)$ denotes the ELU activation function, and $[\cdot;\cdot]$ represents feature concatenation. The skip connections for layers $i > 3$ allow activations from three layers prior to influencing the current layer, which helps in feature reuse while maintaining the original input's accessibility. Each layer's output is concatenated with the physical-terrain representation, as shown below:

$$D^{(i)} = [Z; H^{(i)}]. \tag{11}$$

This structure preserves the fundamental atmospheric dynamics throughout the transformations, ensuring that key features from both the physics and terrain models are retained. The output weights $\zeta_i \in \mathbb{R}^{(2H+d)\times H}$ are obtained by solving the regularized least squares problem:

$$\zeta_i = \left(D^{(i)\top} D^{(i)} + \lambda I\right)^{-1} D^{(i)\top} Y, \tag{12}$$

where Y represents the future temperature values. This regularized least squares formulation ensures a gradient-free optimization process, which is computationally efficient compared to gradient-based methods. The final fused prediction is obtained by aggregating the outputs from all layers using median aggregation:

$$y_{\text{fused}} = \text{median}\left(D^{(1)}\zeta_1, D^{(2)}\zeta_2, \dots, D^{(L)}\zeta_L\right). \tag{13}$$

This median-based aggregation enhances the robustness of the fusion mechanism, making it less sensitive to initialization variability. The fusion process captures nonlinear interactions between the physical model and terrain effects through random feature expansions, while the skip connections ensure that important features persist across the network layers. Furthermore, the ensemble aggregation method provides stability to the predictions by leveraging the diversity of the multiple layers. Empirical results demonstrate that the fusion of physical and terrain-based predictions via the ed-RVFL-SC network improves mean absolute percentage error compared to using each submodule independently, as observed in the ablation study on the Calgary dataset. Moreover, on the Tucman dataset, combining both models improves all evaluation metrics, outperforming the individual models. This improvement arises from the fusion's ability to correct systematic biases in the physics-based model while also incorporating terrain-mediated influences, providing more accurate forecasts in some cases.

3.2. Spatio-temporal convolutional LSTM agent

The Spatio-Temporal Convolutional LSTM (STCL) agent is the core component of the network, responsible for handling the prediction tasks by learning both spatial and temporal dependencies in the data. The training process occurs in two stages, each using a distinct loss function: Mean Absolute Error (MAE) and RAFL. This section describes the first stage, which utilizes MAE loss to minimize the overall absolute prediction error. Fig. 2, shows the block diagram of the proposed STCL agent.

Initially, the input X' is reshaped by adding an extra dimension, making it compatible with the 3D convolutional layer. This reshaping operation ensures that the input tensor $X' \in \mathbb{R}^{B\times T\times F\times\theta\times 1}$ correctly matches the expected dimensions for the convolutional

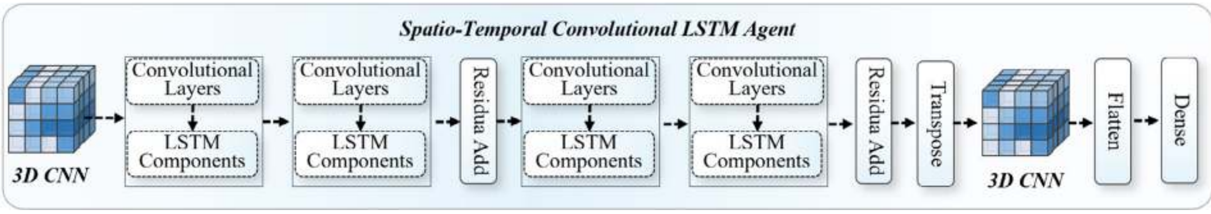

Fig. 2. Block diagram of the proposed spatio-temporal convolutional LSTM agent.

operations. The reshaped input tensor X' is passed through a 3D convolutional layer designed to capture the temporal dependencies across all features. The convolution operation is given as:

$$\mathbf{Z}(b,t,f,\theta,c) = \sum_{m=0}^{M-1}\sum_{i=0}^{I-1}\sum_{j=0}^{J-1}\sum_{k=0}^{K-1} \mathbf{X}'(b,t+i,f+j,\theta+k,c)\cdot W(i,j,k,c). \tag{14}$$

where, $\mathbf{X}'(b,t+i,f+j,\theta+k,c)$ represents the input tensor at position $(b,t+i,f+j,\theta+k,c)$, and the kernel $W(i,j,k,c)$ performs the convolution over the spatial and feature locations. This convolutional operation is key in extracting the temporal dependencies for each spatial location and feature. The output from the 3D convolutional layer is then passed through two ConvLSTM layers. The first ConvLSTM layer reduces the number of filters by half, and the second layer restores the original filter count. This process allows the network to compactly represent the temporal features in the first layer and then expand them for detailed temporal analysis in the second layer. The ConvLSTM units integrate convolutions and LSTM gates to model both spatial and temporal dependencies efficiently. The ConvLSTM operations are defined by the following equations:

$$f_t = \sigma(W_f * \mathbf{Z}_t + U_f * h_{t-1} + b_f), \tag{15}$$

$$i_t = \sigma(W_i * \mathbf{Z}_t + U_i * h_{t-1} + b_i), \tag{16}$$

$$\tilde{c}_t = \tanh(W_c * \mathbf{Z}_t + U_c * h_{t-1} + b_c), \tag{17}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{18}$$

$$o_t = \sigma(W_o * \mathbf{Z}_t + U_o * h_{t-1} + b_o), \tag{19}$$

$$h_t = o_t \odot \tanh(c_t). \tag{20}$$

where $\mathbf{Z}_t$ is the input at time step t, h_{t-1} is the previous hidden state, and c_{t-1} is the previous cell state. The σ and $\tanh$ functions are the sigmoid and hyperbolic tangent activation functions, respectively, and the weight matrices W_f, W_i, W_c, W_o and U_f, U_i, U_c, U_o are learnable parameters, while b_f, b_i, b_c, b_o are bias terms. After processing the input through ConvLSTM layers, the learned temporal features h_t are added back to the original spatial features to mitigate the vanishing gradient problem and retain both the spatial and temporal dependencies. This operation is mathematically expressed as:

$$\mathbf{X}'' = \mathbf{X}' + h_t. \tag{21}$$

The combination of these learned features ensures the network's ability to capture and retain both the spatial and temporal aspects of the input. After passing through multiple ConvLSTM layers, the final tensor is transposed so that the time dimension is placed at the last axis. The tensor is then processed through a final 3D convolution layer to extract both temporal and spatial features. The output tensor is flattened and passed through a dense layer with H units, where H corresponds to the prediction horizon. The final prediction output, denoted as $Y_{\text{STCL\_baseline}}$, represents the model's time-series forecast, incorporating both short- and long-term dependencies. The STCL Agent effectively models both temporal and spatial dependencies through a combination of 3D convolutions and ConvLSTM layers. This architecture enables the model to learn complex temporal dynamics while maintaining computational efficiency, making it well-suited for time-series forecasting tasks.

3.3. Regime-Specialized agent

The Regime-Specialized Agent emerges from observed limitations in the Physics-Terrain Fusion Agent's performance during extreme temperature events. When y_{fused} encounters significant prediction errors, particularly during extreme temperatures. Such cases have sparse examples therefore the existing MAE trained model also fails in such cases as it tries to minimize average error. Hence, there is a need for a data model that is trained where the previous two agents are failing.

The rationale for retraining the existing STCL baseline agent, rather than constructing a separate architecture, is threefold. First, it ensures computational and parametric efficiency by building upon the already learned spatio-temporal feature representations, avoiding the cost of training a new large model from scratch. Second, this approach promotes stability and knowledge transfer; the baseline agent has already captured the dominant meteorological patterns, allowing the RAFL to specialize this capable model by fine-tuning its attention toward the previously problematic residual errors. Finally, and most critically, it creates a complementary expert system. A separate model might learn to make predictions independently, potentially diverging from or even contradicting the baseline. By retraining the original architecture, the regime-specialized agent learns to specifically correct the baseline's mistakes on hard examples, resulting in a more cohesive and collaborative ensemble where each component addresses a distinct aspect of the prediction challenge.

To quantify these error patterns, this work develops a composite metric combining RMSE's sensitivity to outliers with MAE's robustness,

$$E_i = \frac{1}{2}\left(\sqrt{\frac{1}{H}\sum_{t=1}^{H}(y^{(i)}_{\text{true},t} - y^{(i)}_{\text{fused},t})^2} + \frac{1}{H}\sum_{t=1}^{H}|y^{(i)}_{\text{true},t} - y^{(i)}_{\text{fused},t}|\right). \tag{22}$$

where H represents the prediction horizon. This dual formulation ensures both sporadic extremes and consistent biases receive appropriate emphasis, unlike single-metric approaches that might overlook one error characteristic. Recognizing that meteorological regimes exhibit gradual transitions rather than discrete boundaries, we implement fuzzy classification. The method calculates global error statistics,

$$E_{\text{min}} = \min_i E_i, \quad E_{\text{max}} = \max_i E_i, \quad c_m = \frac{E_{\text{min}} + E_{\text{max}}}{2}. \tag{23}$$

Then assigns continuous membership values through triangular functions,

$$\mu_{\text{low}}(E) = \max\left(0, \frac{c_m - E}{c_m - E_{\text{min}}}\right), \tag{24}$$

$$\mu_{\text{medium}}(E) = \max\left(0, \min\left(\frac{E - E_{\text{min}}}{c_m - E_{\text{min}}}, \frac{E_{\text{max}} - E}{E_{\text{max}} - c_m}\right)\right), \tag{25}$$

$$\mu_{\text{high}}(E) = \max\left(0, \frac{E - c_m}{E_{\text{max}} - c_m}\right) \tag{26}$$

producing membership vectors $M^{(i)} = [\mu_{\text{low}}(E_i), \mu_{\text{medium}}(E_i), \mu_{\text{high}}(E_i)]$. This soft partitioning accommodates transitional weather states where binary classification would introduce discontinuities. Training uses RAFL that addresses three interconnected challenges, example importance weighting, gradient stability and persistent error correction. The loss first computes the base prediction error,

$$\mathcal{L}^{(i)}_{\text{MSE}} = \frac{1}{H}\sum_{t=1}^{H}\left(y^{(i)}_{\text{true},t} - y^{(i)}_{\text{STCL\_RAFL},t}\right)^2, \tag{27}$$

then incorporates regime membership through adaptive weighting,

$$w^{(i)} = \alpha M^{(i)}_1 + \beta M^{(i)}_2 + \gamma M^{(i)}_3 \quad \text{with} \quad \gamma > \beta > \alpha, \tag{28}$$

where α, β, γ control regime emphasis. To prevent gradient explosion during extreme events, a stability term combines current and historical errors,

$$\mathcal{L}^{(i)}_{\text{stability}} = \mathcal{L}^{(i)}_{\text{MSE}} + \eta E_i, \tag{29}$$

with η scaling historical influence. The focal component then intensifies focus on stubborn mispredictions,

$$c^{(i)} = \exp\left(-\text{clip}(\mathcal{L}^{(i)}_{\text{stability}}, -50, 50)\right), \quad \mathcal{L}^{(i)}_{\text{focal}} = \left(1 - \text{clip}(c^{(i)}, \epsilon, 1-\epsilon)\right)^{\delta}, \tag{30}$$

where δ modulates focusing intensity. The integrated loss becomes,

$$\mathcal{L} = \frac{1}{B}\sum_{i=1}^{B}\frac{w^{(i)}}{\max_j w^{(j)} + \epsilon} \cdot \mathcal{L}^{(i)}_{\text{focal}} \cdot \mathcal{L}^{(i)}_{\text{MSE}}, \tag{31}$$

with B denoting batch size. This formulation ensures high-error regimes receive dominant weighting while adaptively hardening focus on recurring errors, creating a self-correcting specialization mechanism.

The agent shares the spatio-temporal architecture from Section 3.2 but trains exclusively on the RAFL. Crucially, the Physics-Terrain Fusion Agent's parameters remain frozen during optimization, forcing the specialized agent to develop complementary capabilities without degrading foundational physics. This focused training produces an expert component, denoted as $Y_{\text{STCL\_RAFL}}$, which significantly enhances prediction accuracy during the most challenging meteorological conditions.

3.4. *Hierarchical adaptive fusion framework*

The hierarchical fusion integrates predictions from all three agents through the ensemble deep random vector functional link with skip connections (edRVFL-SC) architecture defined in Section 3.1.3. The input matrix $Z_{\text{fusion}} = [y_{\text{fused}}, y_{\text{STCL\_baseline}}, y_{\text{STCL\_RVFL}}] \in \mathbb{R}^{n\times 3H}$ concatenates outputs from the Physics-Terrain, Spatio-Temporal and Regime-Specialized agents, expanding the original $2H$-dimensional input from Eq. (16) to accommodate the additional agent. The network processes Z_{fusion} through identical hierarchical

Table 1
Statistical summary of hourly temperature at 2 m height and data partitioning.

Dataset	Mean	Median	Std Dev	Min	Max	Variance	Data partition (h)		
	(°C)	(°C)	(°C)	(°C)	(°C)	(°C^2)	Train	Valid	Test
Beijing	12.04	12.91	12.65	−21.93	43.33	160.02	61 370	13 151	13 151
Calgary	3.92	3.43	11.30	−39.49	34.90	127.69	61 370	13 151	13 151
Dhaka	25.51	26.61	5.44	6.54	43.27	29.59	61 370	13 151	13 151
Tucumán	22.01	22.14	7.75	−0.43	45.60	60.06	61 370	13 151	13 151

Note: Temperature values are in degrees Celsius (°C), variance in square degrees Celsius (°C^2). All datasets use the same partitioning scheme, where the first 61,370 h are used for training all models, the next 13,151 h are used for validation, and the last 13,151 h, which have never been shown during training or validation, are reserved for testing to prevent data leakage.

layer computations (Eq. (16)), skip connection structures (Eq. (17)) and output weight solutions (Eq. (18)) as the Physics-Terrain fusion, maintaining the L-layer architecture with ELU activations and random weight initialization. Final predictions emerge through the same median aggregation mechanism (Eq. (19)), preserving the stability of the ensemble and the robustness of initialization proven in Section 3.1.3. This direct architectural reuse leverages the edRVFL-SC's established capacity for modeling nonlinear interactions while maintaining information persistence through skip connections, now dynamically balancing physics-constrained foundations, data-driven patterns and regime-specialized corrections within a unified prediction framework. The only modification is in increasing the input dimension from $2H$ to $3H$ maintains computational efficiency while allowing the integration of the entire agent.

4. Results and comparisons

We evaluate PIEL-NET across four cities with diverse climates and report both average and tail-focused metrics, plus ablations isolating each agent's contribution. The evaluation includes both a comparison of model performance across datasets and a detailed analysis of individual model components through an ablation study. Additionally, PIEL-NET is benchmarked against state-of-the-art (SOTA) models to assess its relative efficacy in capturing temperature dynamics across different environments. The datasets used in this study span a range of geographic and climatic conditions, providing a comprehensive test of the model's robustness and generalizability.

4.1. Spatio-temporal dataset for temperature prediction

This work evaluates PIEL-NET using hourly temperature data from four globally distributed urban regions, namely Beijing (China), Calgary (Canada), Dhaka (Bangladesh), and Tucumán (Argentina). Each city is represented by the reanalysis grid cell containing its center, with additional data from surrounding locations at distances of 100 km in each cardinal and inter-cardinal direction. This 100 km offset ensures that the model is provided not only with city-center data but also with data from nearby, geographically distinct regions. These surrounding locations provide a broader regional perspective on temperature dynamics, capturing variations influenced by surrounding terrains, urban heat islands, and regional climatic conditions. By including data from these offset locations, the model gains insights into the spatial factors that affect temperature predictions, improving its ability to generalize across diverse environments.

This study predicts the *city-center grid cell* temperature, defined as the NASA POWER 2 m air temperature (T2M) average of the reanalysis grid cell centered on the administrative urban core, at nominal $0.5° \times 0.625°$ resolution (≈50–70 km depending on latitude). This resolution aggregates intra-urban heterogeneity and does not resolve neighborhood microclimates or street-canyon effects. Therefore we interpret all results at the city-center grid cell rather than the street scale. For transparency, we report for each site the overlap between the mapped urban extent and the central grid cell and note cases where a municipality spans multiple cells. The nine-point sampling with 100 km radial offsets supplies an advective context that influences the central cell and supports computationally efficient, cross-city-comparable early warning at the city-center grid cell.

The spatio-temporal nature of the data is critical to this study. For each urban area, hourly temperature data is collected from the central grid cell (representing the city center) as well as from eight peripheral locations situated 100 km away in the four cardinal and four inter-cardinal directions. Along with temperature, the dataset also includes humidity, wind speed, wind direction, and air pressure data, measured at each of these locations. This configuration allows the model to account for both temporal changes in temperature over time (with hourly samples) and spatial variations within the region. The combination of data from multiple geographically diverse locations creates a comprehensive dataset that enables the model to learn patterns across both space and time.

The dataset used in this study originates from NASA's POWER project (Stackhouse et al., 2015) and spans a period of 10 years (2005–2014). It includes hourly temperature measurements from the city center and the eight peripheral locations. Table 1 summarizes key temperature statistics for each city. Beijing exhibits the largest annual range in temperature (65.3 °C), Dhaka maintains tropical stability (variance of 29.59 °C), Calgary records extreme minima (−39.49 °C), and Tucumán shows significant seasonal transitions, with the highest temperature variance observed in September.

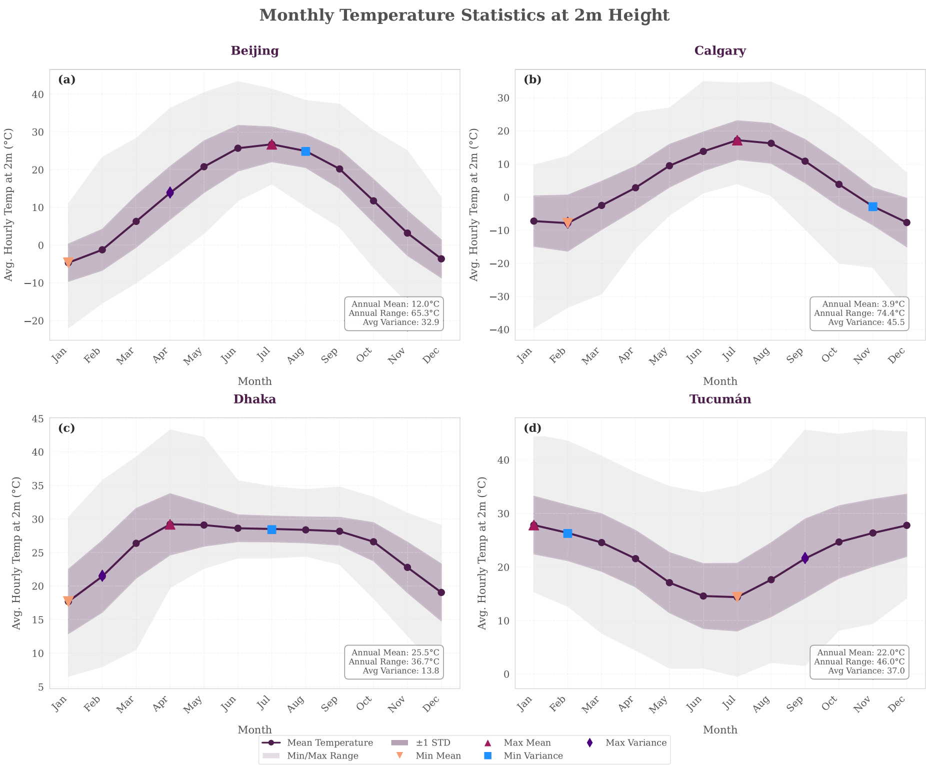

Fig. 3. Monthly temperature characteristics showing (a) Beijing's continental extremes, (b) Calgary's high-latitude variability, (c) Dhaka's tropical consistency, and (d) Tucumán's subtropical seasonality. Shaded regions indicate ±1 STD, while markers denote extreme months.

Fig. 3 visualizes these climatic differences through monthly profiles. Beijing demonstrates sharp summer-winter transitions with July peaks (26.7 °C) and January troughs (−4.7 °C). Calgary exhibits greater variance during winter months (February variance ≈71.9, i.e., STD ≈ 8.5 °C) due to continental air mass collisions. Dhaka maintains stable temperatures year-round with minimal fluctuation (July STD = 3.7). Tucumán shows asymmetric transitions with rapid spring warming and gradual autumn cooling. This diverse selection of cities allows for the evaluation of the model's performance across hemispheric contrasts (Northern and Southern seasons), thermal extremes (ranging from arctic to tropical climates), topographic diversity (including coastal, mountain, and plains regions), and variance patterns (comparing low-variance tropical climates with high-variance continental climates). Identical data partitioning (61,370 train/13,151 validation/13,151 test hours) ensures standardized comparison against state-of-the-art approaches. The spatial configuration (city center + 8 directions) captures advection effects critical for city-center grid cell temperature prediction.

4.2. Experimental setup

The experimental framework is designed to evaluate the performance of the proposed PIEL-NET model compared to existing models under standard conditions. Training runs for 1000 epochs with early stopping and a patience of 10 epochs. The Adam optimizer is used, and MAE loss is applied for training all models, except for the Regime-Specialized Agent, which uses RAFL as the loss function. The model takes 48 h of past data as input with 49 features to predict the temperature for the next 12 h. The batch size is set to 128. Training is conducted on an NVIDIA RTX 3080 Ti GPU, and inference is performed on the test subset. Key parameters are summarized in Table 2.

4.2.1. Model hyperparameter optimization and external validity

We tune the hyperparameter configuration ϑ for each model one time on the Beijing data using Tree-structured Parzen Estimation with an equal trial budget. For every other city s, we reinitialize all learnable weights w_s for every model and train them from scratch on the local training split while keeping ϑ fixed. No fitted weights or batch statistics transfer across the different city sites for any model. We also recompute the site-specific normalization and any model-specific parameters, such as the physics coefficients (α, β) for PIEL-NET or the Regime-Adaptive Focal Loss statistics, using only local data to prevent information leakage. We select Beijing as the development city because its data in Table 1 shows the widest dispersion and strong warm tails. This reduces the risk that ϑ will be over-specialized to a simpler climate. We then report test results for each city using the fixed ϑ and freshly trained weights w_s for all models. This strategy evaluates the transfer of each model's inductive bias while preserving adaptation to local conditions, providing a more rigorous and fair test of operational reproducibility and generalizability.

Table 2
Key experimental parameters.

Parameter	Value
Training epochs	1000
Early stopping patience	10
Optimizer	Adam
Loss function	MAE and RAFL
Input data	48 h (49 features)
Prediction horizon	12 h
Batch size	128
Training and testing hardware	NVIDIA RTX 3080

4.3. Comparison metrics

To evaluate the performance of the proposed PIEL-NET model against state-of-the-art (SOTA) models, we employ Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R-squared Score), Mean Absolute Percentage Error (MAPE), Weighted Absolute Percentage Error (WAPE) and Symmetric Mean Absolute Percentage Error (SMAPE). These metrics collectively assess predictive accuracy, robustness and generalizability in TP. RMSE quantifies the square root of the average squared differences between predicted and actual temperatures,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_{\mathrm{true}}^{(i)} - y_{\mathrm{pred}}^{(i)})^2} \tag{32}$$

where $y_{\mathrm{true}}^{(i)}$ denotes the actual temperature, $y_{\mathrm{pred}}^{(i)}$ the predicted temperature and N the number of predictions. RMSE emphasizes larger errors due to the squaring operation, making it sensitive to extreme deviations. In TP, RMSE highlights model performance during significant weather events, such as heatwaves or cold snaps, where large errors impact reliability. MAE computes the average absolute differences between predicted and actual temperatures,

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_{\mathrm{true}}^{(i)} - y_{\mathrm{pred}}^{(i)}| \tag{33}$$

MAE treats all errors equally, providing a straightforward measure of typical error magnitude. For TP, MAE indicates the average deviation across all conditions, offering a balanced view of model consistency. R-squared Score measures the proportion of variance in actual temperatures explained by the model,

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{N}(y_{\mathrm{true}}^{(i)} - y_{\mathrm{pred}}^{(i)})^2}{\sum_{i=1}^{N}(y_{\mathrm{true}}^{(i)} - \bar{y}_{\mathrm{true}})^2} \tag{34}$$

where $\bar{y}_{\mathrm{true}}$ is the mean of actual temperatures. Values range from 0 to 1, with 1 indicating a perfect fit. In TP, a high R-squared Score confirms the model's ability to capture underlying climatic patterns and variability across diverse regions. MAPE calculates the average percentage difference between predicted and actual temperatures,

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_{\mathrm{true}}^{(i)} - y_{\mathrm{pred}}^{(i)}}{y_{\mathrm{true}}^{(i)}}\right| \times 100 \tag{35}$$

MAPE expresses errors relative to actual values, enabling comparison across datasets with different temperature scales. In TP, MAPE facilitates the evaluation of proportional accuracy, critical for assessing performance in varied climatic conditions like tropical or continental climates. WAPE weights errors by actual temperature values,

$$\mathrm{WAPE} = \frac{\sum_{i=1}^{N}|y_{\mathrm{true}}^{(i)} - y_{\mathrm{pred}}^{(i)}|}{\sum_{i=1}^{N}|y_{\mathrm{true}}^{(i)}|} \times 100 \tag{36}$$

WAPE prioritizes errors at higher temperatures, which often carry greater practical significance. In TP, WAPE emphasizes accuracy during critical events, such as extreme heat, where precise forecasts impact decision-making. SMAPE modifies MAPE to symmetrize error calculation,

$$\mathrm{SMAPE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_{\mathrm{true}}^{(i)} - y_{\mathrm{pred}}^{(i)}|}{|y_{\mathrm{true}}^{(i)}| + |y_{\mathrm{pred}}^{(i)}|} \times 100 \tag{37}$$

SMAPE averages predicted and actual values in the denominator, avoiding undefined results when actual temperatures approach zero, a concern in cold climates. For TP, SMAPE ensures consistent evaluation across all temperature ranges, enhancing comparability in regions with extreme lows. This set of metrics, including RMSE, MAE, R-squared Score, MAPE, WAPE and SMAPE, enables a comprehensive comparison of PIEL-NET with SOTA models, addressing diverse aspects of TP performance.

Table 3
Overview of model versions and added components.

Model	Wind-driven heat transfer modeling	Terrain-aware feature interaction	Fusion (edRVFL-SC)	Spatio-temporal convolutional agent	Regime-specialized agent	Final fusion
PIEL-NET-V1	✓					
PIEL-NET-V2		✓				
PIEL-NET-V3	✓	✓	✓			
PIEL-NET-V4				✓		
PIEL-NET-V5					✓	
PIEL-NET-V6	✓	✓	✓	✓	✓	✓

Table 4
Average performance metrics for all model versions on the Beijing data.

Model	RMSE (°C)	MAE (°C)	MAPE (%)	WAPE (%)	SMAPE (%)	R-squared score
PIEL-NET-V1	4.0711	2.9572	97.60%	19.53%	42.27%	0.8930
PIEL-NET-V2	1.6704	1.1970	39.46%	7.91%	20.46%	0.9822
PIEL-NET-V3	2.9605	2.4204	50.51%	15.99%	30.45%	0.9449
PIEL-NET-V4	1.6244	1.1902	40.91%	7.86%	20.84%	0.9833
PIEL-NET-V5	2.2626	1.7852	61.41%	11.79%	28.42%	0.9678
PIEL-NET-V6	1.5575	1.1080	35.34%	7.32%	19.14%	0.9845

Furthermore, to evaluate the performance of the early warning system for extreme temperature events, we employ four key classification metrics. These metrics assess the system's capability to reliably detect thermal extremes while minimizing false alarms and missed detections. Precision quantifies the proportion of correct warnings among all issued alerts

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{38}$$

where TP represents true positives (correct extreme event warnings) and FP denotes false positives (false alarms during normal conditions). High precision indicates minimal false alarms, which is crucial for maintaining operational credibility. Recall measures the system's ability to detect actual extreme events

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{39}$$

where FN indicates false negatives (missed extreme events). High recall ensures comprehensive coverage of dangerous temperature conditions. Accuracy represents the overall correctness of the warning system

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{40}$$

where TN signifies true negatives (correct non-warning decisions). The Dice coefficient balances precision and recall

$$\text{Dice} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}} \tag{41}$$

This metric proves particularly valuable for operational early warning systems, where false alarms and missed detections carry significant consequences. These metrics enable a rigorous evaluation of the operational effectiveness of the warning system for urban climate risk management.

4.4. Ablation study

This section presents an ablation study to evaluate the contribution of different components in the proposed PIEL-NET model. PIEL-NET consists of multiple agents, each having its own contribution to enhance the accuracy of TP. The model's Wind-Driven Heat Transfer Modeling is represented as PIEL-NET-V1, while its terrain-aware feature interaction is encapsulated in PIEL-NET-V2. The fusion of these two components is achieved in PIEL-NET-V3, which incorporates the edRVFL-SC. Beside this an alternative approach is represented by PIEL-NET-V4, which integrates the STCL agent trained with MAE loss. PIEL-NET-V5 introduces a regime-specialized agent for tackling difficult prediction scenarios that is trained with RAFL loss. The final model, PIEL-NET-V6, combines the outputs of PIEL-NET-V3, PIEL-NET-V4 and PIEL-NET-V5 to final output of the proposed model. The overview of model versions and added components is given in Table 3. The following sub-sections present detailed evaluations of these model versions across different datasets, allowing for a comparison of their performance and the contribution of each component to the model's overall effectiveness.

4.4.1. Ablation study on Beijing data

This section discusses the ablation study to show systematic improvement in performance on the Beijing data. PIEL-NET-V1, based on isolated Wind-Driven Heat Transfer Modeling, shows high prediction errors (RMSE = 4.0711, MAE = 2.9572) due to difficulties in capturing spatio-temporal dynamics for temperature prediction. PIEL-NET-V2, which incorporates terrain-sensitive

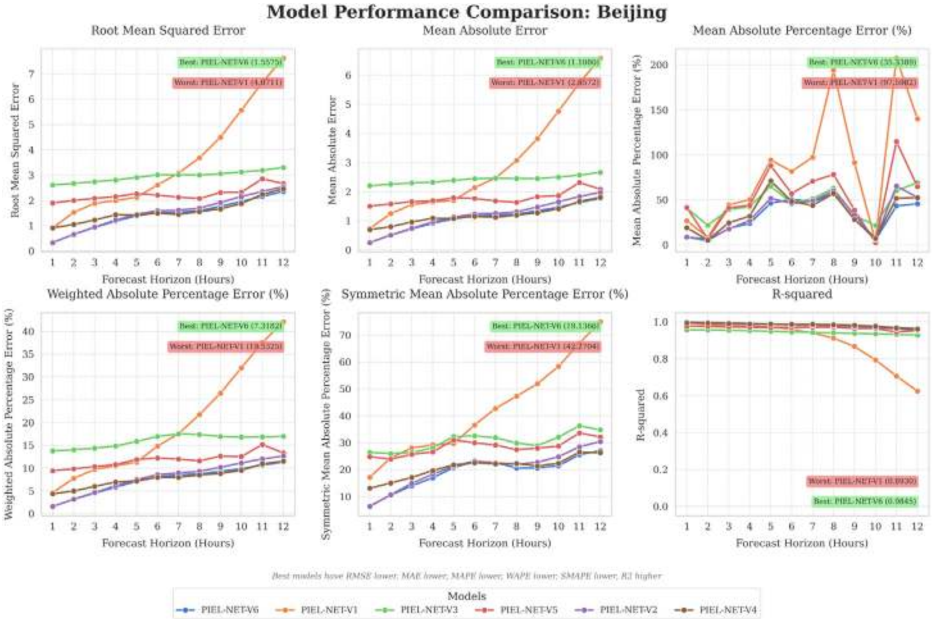

Fig. 4. Metrics hourly comparison across PIEL-NET variants for Beijing data.

Table 5
Overall performance metrics for all model versions on the Calgary dataset.

Model	RMSE (°C)	MAE (°C)	MAPE (%)	WAPE (%)	SMAPE (%)	R-squared score
PIEL-NET-V1	4.0017	3.3579	158.17	34.63	60.79	0.8097
PIEL-NET-V2	1.7775	1.4005	80.02	14.47	33.18	0.9685
PIEL-NET-V3	1.8773	1.4859	72.43	15.34	35.59	0.9674
PIEL-NET-V4	1.6935	1.2928	70.50	13.35	31.29	0.9740
PIEL-NET-V5	2.5895	2.0741	118.01	21.37	43.11	0.9441
PIEL-NET-V6	1.6089	1.2484	68.15	12.90	30.88	0.9746

feature interactions, reduces MAE by 59.5%, confirming the effectiveness of the module. PIEL-NET-V3, which combines Physics-Terrain Fusion Agent, performs worse than V2, with a 50.51% MAPE, possibly due to conflicting signals from the fusion of physics and terrain features. PIEL-NET-V4, using a spatio-temporal convolutional LSTM agent, achieves a lower MAE (1.1902) but shows 40.91% MAPE during temperature extremes, revealing limitations in capturing high-impact events. PIEL-NET-V5, using RAFL, reduces MAE by 26.2% compared to V3 but still has 61.41% MAPE, indicating issues with isolating pure regimes during rapid weather transitions. PIEL-NET-V6, incorporating a hierarchical fusion framework, reduces MAE by 54.2% from V2 and shows the most improvement during critical transition periods, with a 41.7% reduction in extreme errors compared to V4. These results show that integrating components through adaptive fusion improves performance, effectively addressing Beijing's complex atmospheric conditions. Table 4 summarizes the average results of all versions for Beijing data.

Detail hourly comparison of all metrics is shown in Fig. 4. The physics-based model PIEL-NET-V1 outperforms PIEL-NET-V3 and PIEL-NET-V5 across nearly all metrics. However, its error increases exponentially with longer prediction horizons. Notably, while PIEL-NET-V2 surpasses the final version (PIEL-NET-V6) in RMSE, MAE and WAPE for horizons up to 12 h, PIEL-NET-V6 dominates all variants comprehensively across all metrics. This shows that systematic refinement yields substantial performance gains.

4.4.2. Ablation study on Calgary dataset

Table 5 reveals distinct performance patterns across model variants. The physics-only model (V1) shows the highest errors (MAE = 3.3579 °C, MAPE = 158.17%), while the terrain-aware model (V2) achieves significant improvement (MAE = 1.4005 °C, MAPE = 80.02%). The combined Physics-Terrain Fusion Agent (V3) exhibits a 6.1% MAE increase (1.4859 °C) compared to V2 but demonstrates a 9.5% MAPE reduction (72.43% vs. 80.02% in V2). This MAPE improvement indicates enhanced performance during large-magnitude temperature variations despite higher absolute errors. The Spatio-Temporal Convolutional LSTM Agent (V4) outperforms V3 across all metrics (MAE = 1.2928 °C, MAPE = 70.50%), while the regime-specialized agent (V5) shows degraded results (MAE = 2.0741 °C, MAPE = 118.01%). The full hierarchical fusion (V6) achieves optimal performance (MAE = 1.2484 °C, MAPE = 68.15%), reducing V4's MAE by 3.4% and V2's MAPE by 14.8%. Fig. 5 illustrates these patterns through hourly error profiles.

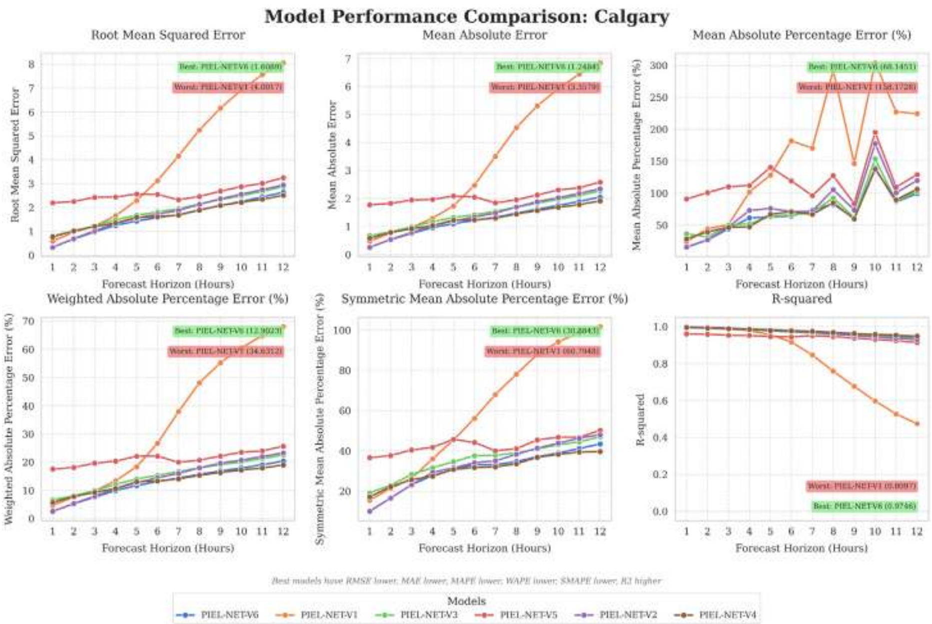

Fig. 5. Metrics hourly comparison across PIEL-NET variants for Calgary dataset.

Table 6
Overall performance metrics for all model versions on the Dhaka dataset.

Model	RMSE (°C)	MAE (°C)	MAPE (%)	WAPE (%)	SMAPE (%)	R-squared score
PIEL-NET-V1	2.4384	1.9266	8.68	7.57	8.08	0.6811
PIEL-NET-V2	0.7833	0.6005	2.53	2.36	2.52	0.9725
PIEL-NET-V3	1.0951	0.8460	3.63	3.31	3.66	0.9573
PIEL-NET-V4	0.7831	0.5889	2.59	2.31	2.54	0.9763
PIEL-NET-V5	1.9923	1.5522	6.44	6.07	6.65	0.8589
PIEL-NET-V6	0.7312	0.5461	2.32	2.15	2.32	0.9762

4.4.3. Ablation study on Dhaka dataset

Table 6 presents the performance metrics across model variants for the Dhaka dataset. The physics-only model (V1) shows moderate performance (MAE = 1.9266 °C, MAPE = 8.68%), while the terrain-aware model (V2) achieves significant improvement (MAE = 0.6005 °C, MAPE = 2.53%). The Physics-Terrain Fusion Agent (V3) demonstrates a 40.9% MAE increase (0.8460 °C) compared to V2 but maintains better performance than V1 (56.1% MAE reduction). The Spatio-Temporal Convolutional LSTM Agent (V4) shows the best individual performance (MAE = 0.5889 °C, MAPE = 2.59%) among component models, while the regime-specialized agent (V5) underperforms (MAE = 1.5522 °C, MAPE = 6.44%) due to its loss function specifically designed for hard examples. The full hierarchical fusion (V6) achieves optimal results (MAE = 0.5461 °C, MAPE = 2.32%), representing a 7.3% MAE improvement over V4 and a 9.0% MAPE reduction compared to V2. Fig. 6 illustrates these patterns through hourly error profiles.

4.4.4. Ablation study on Tucumán dataset

Table 7 shows performance metrics across model variants for the Tucumán dataset. The physics-only model (V1) exhibits the highest errors (MAE = 2.8748 °C, MAPE = 15.55%). The terrain-aware model (V2) reduces MAE to 1.1411 °C (60.3% improvement) and MAPE to 6.05%. The Physics-Terrain Fusion Agent (V3) achieves MAE = 1.1005 °C (3.6% improvement over V2) and MAPE = 5.73%. The Spatio-Temporal Convolutional LSTM Agent (V4) shows better performance (MAE = 1.0485 °C, MAPE = 5.40%) than V3. The regime-specialized agent (V5) underperforms (MAE = 2.0827 °C, MAPE = 10.70%). The full hierarchical fusion (V6) achieves optimal metrics (MAE = 1.0046 °C, MAPE = 5.15%), representing a 4.2% MAE improvement over V4 and a 14.9% MAPE reduction compared to V2. Fig. 7 illustrates these patterns through hourly error profiles.

The ablation analysis reveals consistent patterns across all urban datasets. The terrain-aware component (V2) generally outperforms the Physics-Terrain Fusion Agent (V3) with marginal advantages in most scenarios. However, V3 demonstrates specific improvements in Tucumán (MAE 1.1005 °C vs. V2 1.1411 °C) and Calgary (MAPE 72.43% vs. V2 80.02%), indicating contextual benefits from physics integration. The regime-specialized agent (V5) systematically underperforms the Spatio-Temporal Convolutional LSTM Agent (V4) across all datasets, consistent with its specialized design focusing on residual errors. The full hierarchical fusion (V6) achieves optimal performance in all cases except Tucumán's RMSE (1.5524 vs. V4 1.4933), while maintaining superior MAE (1.0046 °C vs. 1.0485 °C) and MAPE (5.15% vs. 5.40%). These results validate the complementary

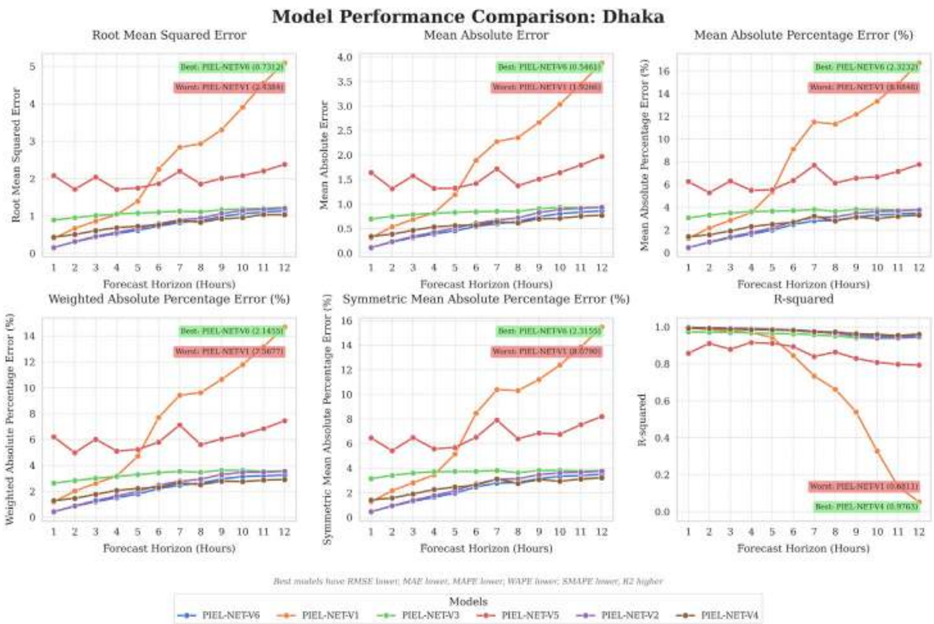

Fig. 6. Metrics hourly comparison across PIEL-NET variants for Dhaka dataset.

Table 7
Overall performance metrics for all model versions on the Tucumán dataset.

Model	RMSE (°C)	MAE (°C)	MAPE (%)	WAPE (%)	SMAPE (%)	R-squared score
PIEL-NET-V1	4.0974	2.8748	15.55	12.50	14.11	0.7033
PIEL-NET-V2	1.6866	1.1411	6.05	4.96	5.85	0.9496
PIEL-NET-V3	1.6413	1.1005	5.73	4.79	5.58	0.9524
PIEL-NET-V4	1.4933	1.0485	5.40	4.56	5.37	0.9615
PIEL-NET-V5	2.6691	2.0827	10.70	9.06	10.34	0.8847
PIEL-NET-V6	1.5524	1.0046	5.15	4.37	5.04	0.9575

Table 8
Performance comparison with SOTA models on the Beijing data.

Model	RMSE (°C)	MAE (°C)	R-squared score	MAPE (%)	WAPE (%)	SMAPE (%)
iTransformer	1.8966	1.4543	0.9771	58.26%	9.63%	25.33%
LSTM	1.8627	1.4436	0.9776	48.14%	9.57%	23.62%
PatchTST	2.2127	1.7201	0.9690	52.93%	11.38%	28.83%
CNN_LSTM	2.2396	1.7427	0.9684	52.53%	11.52%	27.05%
TCN	1.9198	1.5062	0.9765	57.39%	9.98%	25.18%
PIEL-NET	1.5575	1.1080	0.9845	35.34%	7.32%	19.14%

contributions of all components, with V6 reducing MAE by 3.4–10.9% over the best individual agents across diverse meteorological regimes.

4.5. Comparison with state of the art(SOTA) methods

This section compares the performance of PIEL-NET with five SOTA models, including iTransformer (Liu et al., 2023), PatchTST (Nie et al., 2022), CNN-LSTM (Farhangmehr et al., 2025), LSTM (Yang et al., 2022) and TCN (Hewage et al., 2020). These models are selected because they represent common approaches to spatio-temporal predictions. iTransformer and PatchTST use transformers to capture long-range dependencies, essential for modeling meteorological data. CNN-LSTM and LSTM focus on temporal dynamics, with CNNs extracting spatial features and LSTMs handling sequential patterns. TCN uses dilated convolutional layers to address long-range temporal dependencies. These models provide a relevant basis for evaluating the performance of PIEL-NET, particularly in the context of TP across different conditions. Hyper-parameters for all models are optimized using the Beijing data through the Tree-structured Parzen Estimator (Watanabe, 2023) and the optimal parameters are applied in the comparison across all datasets. All baselines use the model selection protocol in Section 4.2.1 to ensure a fair, transfer-focused comparison.

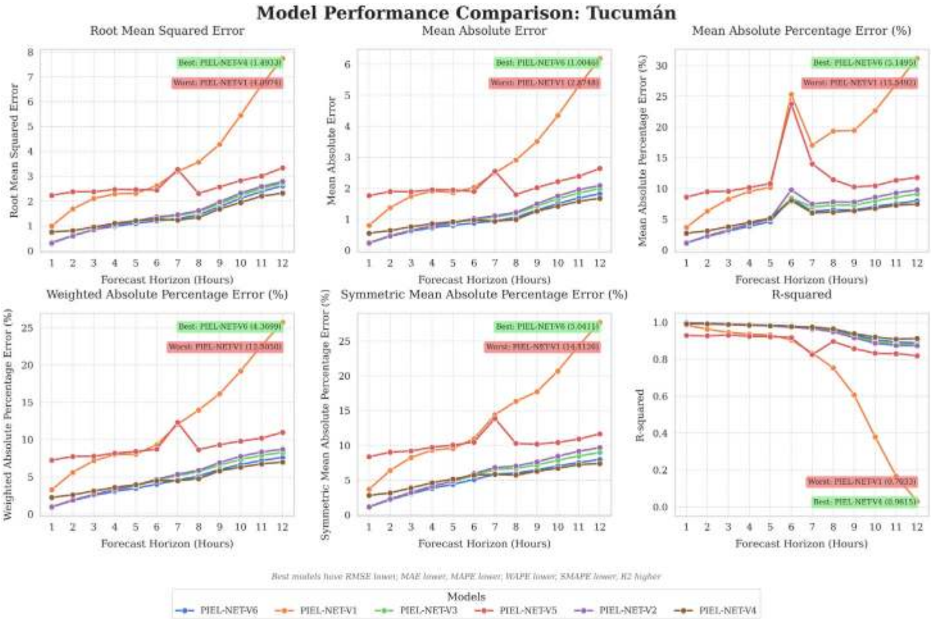

Fig. 7. Metrics hourly comparison across PIEL-NET variants for Tucumán dataset.

4.5.1. Comparison of PIEL-NET with SOTA for the Beijing dataset

Table 8 presents the performance of PIEL-NET and other SOTA models on the Beijing data. A lower RMSE indicates better model performance in capturing outliers or extreme temperature conditions and PIEL-NET achieves the lowest RMSE (1.5575), demonstrating superior performance during such extreme events compared to other models, including iTransformer (1.8966) and TCN (1.9198). For MAE, a lower value reflects better average prediction accuracy across all conditions and PIEL-NET's MAE of 1.1080 is the lowest, indicating that it consistently provides accurate predictions. In contrast, iTransformer and TCN show higher MAE values (1.4543 and 1.5062, respectively), suggesting that their average predictions are less accurate. PIEL-NET also has the highest R-squared Score (0.9845), meaning it explains the largest proportion of variance in the data. The MAPE for PIEL-NET is 35.34% and is the minimal relative error, while other models, such as iTransformer (58.26%) and TCN (57.39%), show much higher relative errors. For WAPE, PIEL-NET's value of 7.32% is the smallest, highlighting its more effective handling of temperature extremes. In terms of SMAPE, PIEL-NET outperforms all models with a value of 19.14%, whereas iTransformer (25.33%) and LSTM (23.62%) show higher symmetric errors. The hourly comparison, shown in Fig. 8, demonstrates PIEL-NET's consistent superior performance across all metrics, particularly during extreme temperature events. Finally, the Wilcoxon signed-rank test results in Fig. 9 indicate that PIEL-NET's performance is significantly different from that of the other models across all metrics, confirming that its improved performance is not due to random chance.

The results presented highlight the effectiveness of PIEL-NET in comparison to other SOTA models on the Beijing data. PIEL-NET demonstrates significant improvements across key performance metrics, including RMSE, MAE, R-squared Score, WAPE and SMAPE, confirming its superior ability to handle both extreme and average temperature conditions. The model's robustness is further substantiated by the hourly comparison in Fig. 8, where PIEL-NET maintains consistent performance across all time horizons. Additionally, the Wilcoxon signed-rank test, shown in Fig. 9, underscores that PIEL-NET's performance is statistically distinct from that of the other models, confirming that the observed improvements are significant and not due to random variations. These findings collectively affirm that PIEL-NET provides a more reliable and effective solution for UTP.

4.5.2. Comparison of PIEL-NET with SOTA for the Calgary dataset

Table 9 presents the performance of PIEL-NET and other SOTA models on the Calgary dataset. A lower RMSE reflects better model performance in capturing extreme temperature variations and PIEL-NET achieves the lowest RMSE (1.6089), showing superior performance during such events compared to models like iTransformer (2.1931) and TCN (2.2910). For MAE, a lower value indicates better average prediction accuracy across all conditions and PIEL-NET's MAE of 1.2484 is the lowest, demonstrating its ability to consistently provide accurate predictions. In contrast, iTransformer and TCN exhibit higher MAE values (1.6771 and 1.7834, respectively), which suggests their predictions are less accurate on average. PIEL-NET also has the highest R-squared Score (0.9746), meaning it explains the largest proportion of variance in the data, indicating better model fit compared to iTransformer (0.9588) and TCN (0.9558). The MAPE for PIEL-NET (68.15%) is significantly lower than those of iTransformer (106.10%) and PatchTST (107.05%), highlighting its superior relative error handling. For WAPE, PIEL-NET's value of 12.90% is the lowest, demonstrating its effective management of temperature extremes. PIEL-NET also outperforms all other models in terms of SMAPE (30.88%), while

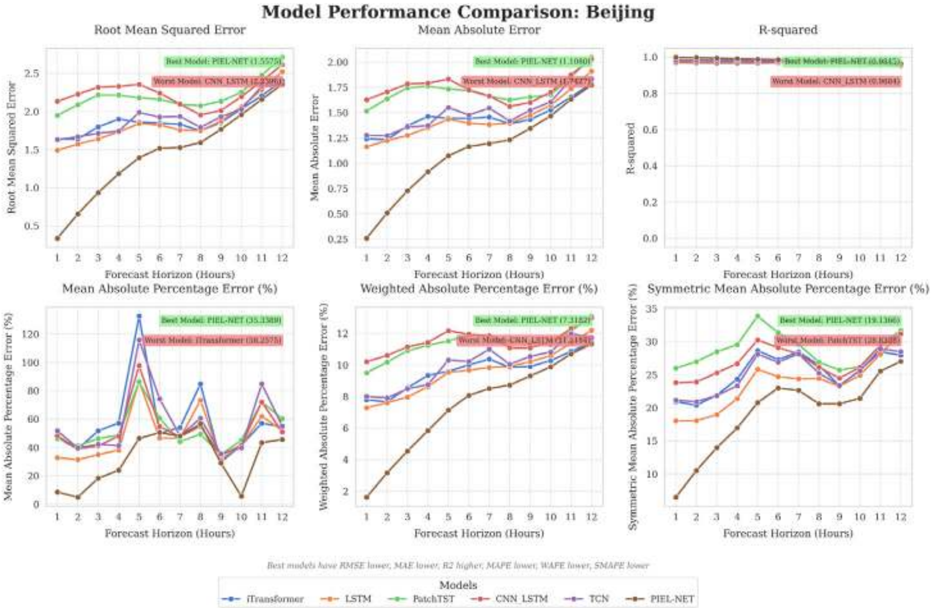

Fig. 8. Hourly comparison of error metrics across PIEL-NET and SOTA models for the Beijing dataset.

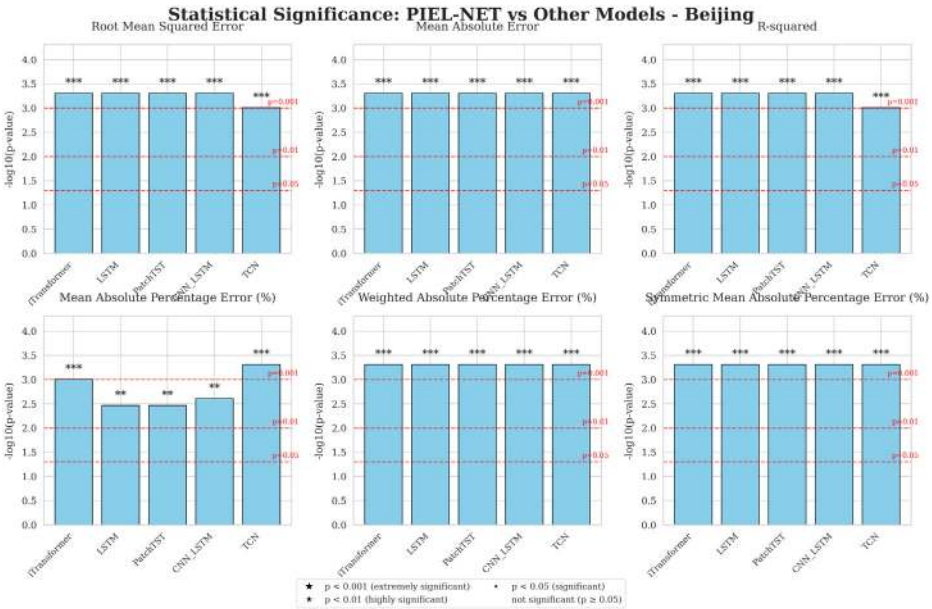

Fig. 9. Wilcoxon signed-rank test results confirming that PIEL-NET is significantly different to all other models.

iTransformer (38.13%) and TCN (40.84%) show higher symmetric errors, indicating less accurate prediction for varying temperature magnitudes.

The hourly comparison, shown in Fig. 10, demonstrates PIEL-NET's superior and consistent performance across all metrics, particularly in capturing extreme temperature fluctuations. Finally, the Wilcoxon signed-rank test results in Fig. 11 indicate that PIEL-NET's performance is statistically different from that of the other models across all metrics, confirming that the improvements are significant and not due to random variations.

4.5.3. Comparison of PIEL-NET with SOTA for the Dhaka dataset

Table 10 provides the performance comparison of PIEL-NET and other SOTA models on the Dhaka dataset. As with the previous datasets, a lower RMSE indicates better performance in capturing extreme temperature variations and PIEL-NET achieves the lowest

Table 9
Performance comparison with SOTA models on the Calgary dataset.

Model	RMSE (°C)	MAE (°C)	R-squared score	MAPE (%)	WAPE (%)	SMAPE (%)
iTransformer	2.1931	1.6771	0.9588	106.10%	17.30%	38.13%
LSTM	2.0213	1.5458	0.9654	74.09%	15.94%	34.86%
PatchTST	3.2485	2.5140	0.9128	107.05%	25.88%	54.90%
CNN_LSTM	2.0415	1.5800	0.9648	89.00%	16.27%	36.68%
TCN	2.2910	1.7834	0.9558	97.71%	18.37%	40.84%
PIEL-NET	1.6089	1.2484	0.9746	68.15%	12.90%	30.88%

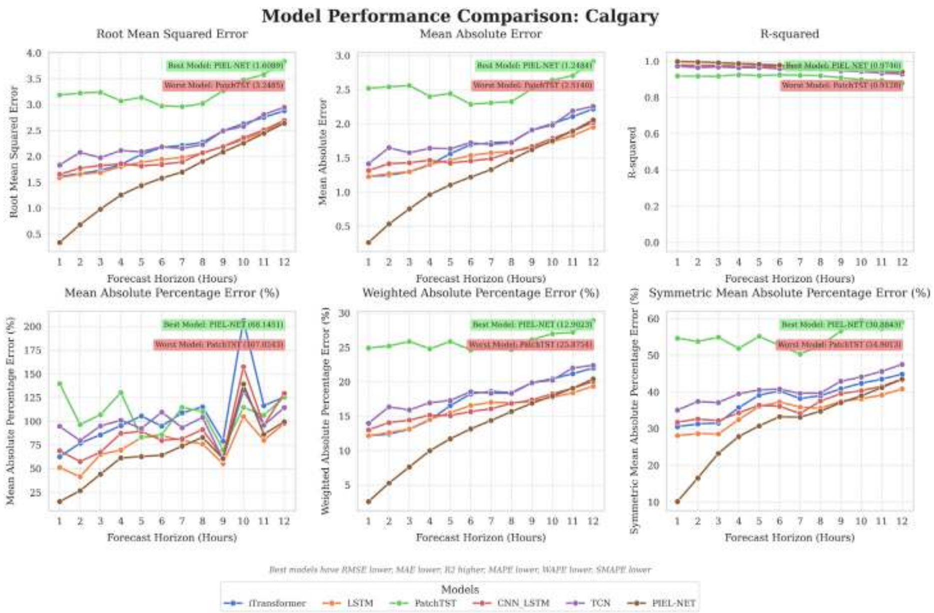

Fig. 10. Hourly comparison of error metrics across PIEL-NET and SOTA models for the Calgary dataset.

Table 10
Performance comparison with SOTA models on the Dhaka dataset.

Model	RMSE (°C)	MAE (°C)	R-squared score	MAPE (%)	WAPE (%)	SMAPE (%)
iTransformer	0.9803	0.7234	0.9663	3.02%	2.83%	3.02%
LSTM	1.9896	1.4835	0.8626	6.25%	5.80%	6.14%
PatchTST	1.1451	0.8443	0.9540	3.51%	3.30%	3.53%
CNN_LSTM	0.9855	0.7494	0.9661	3.10%	2.93%	3.10%
TCN	1.1405	0.8626	0.9545	3.66%	3.37%	3.70%
PIEL-NET	0.7312	0.5461	0.9762	2.32%	2.15%	2.32%

RMSE (0.7312), which highlights its superior accuracy during such events compared to other models like iTransformer (0.9803) and TCN (1.1405). For MAE, which reflects the accuracy of average predictions, PIEL-NET again shows the best result with an MAE of 0.5461, demonstrating consistent performance across all conditions. In comparison, iTransformer (0.7234) and TCN (0.8626) show higher MAE values, suggesting less precise predictions on average. PIEL-NET also leads in R-squared Score (0.9762), which means it explains the largest proportion of variance in the data, outperforming iTransformer (0.9663) and TCN (0.9545). The MAPE of PIEL-NET is 2.32%, lower than iTransformer (3.02%) and TCN (3.66%), highlighting better management of relative errors. For WAPE, PIEL-NET again performs best with a value of 2.15%, reinforcing its effective handling of temperature extremes. Additionally, PIEL-NET outperforms all other models in SMAPE with a value of 2.32%, while iTransformer (3.02%) and TCN (3.70%) exhibit higher symmetric errors, suggesting less accuracy for varying temperature magnitudes.

The hourly comparison in Fig. 12 further illustrates PIEL-NET's robust and consistent performance across all metrics. The Wilcoxon signed-rank test results, shown in Fig. 13, confirm that PIEL-NET's performance is significantly different from that of the other models across all metrics, except for MAPE, where no significant difference is observed between PIEL-NET and LSTM, ensuring that the observed improvements are not attributed to random variations.

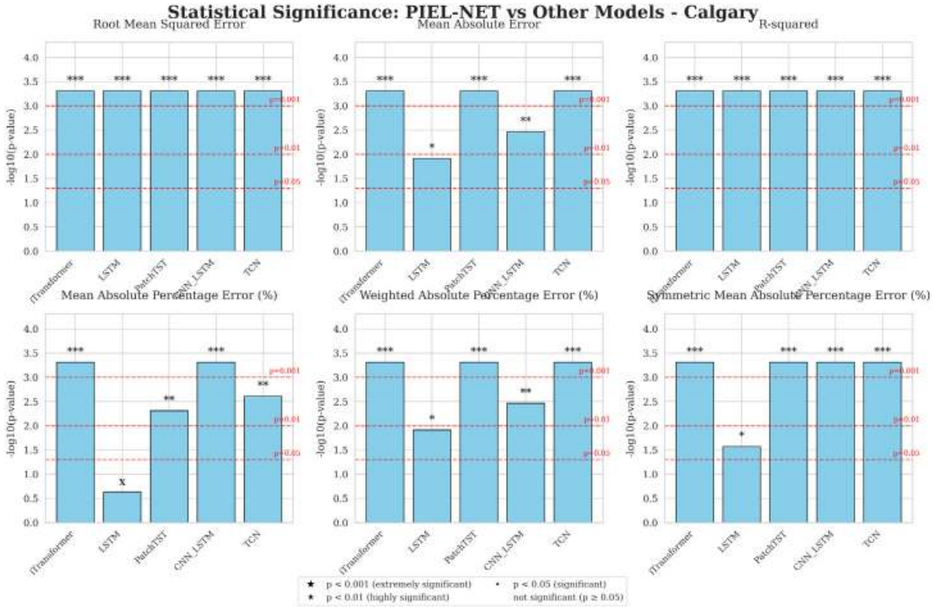

Fig. 11. Wilcoxon signed-rank test results confirming that PIEL-NET is significantly different from all other models.

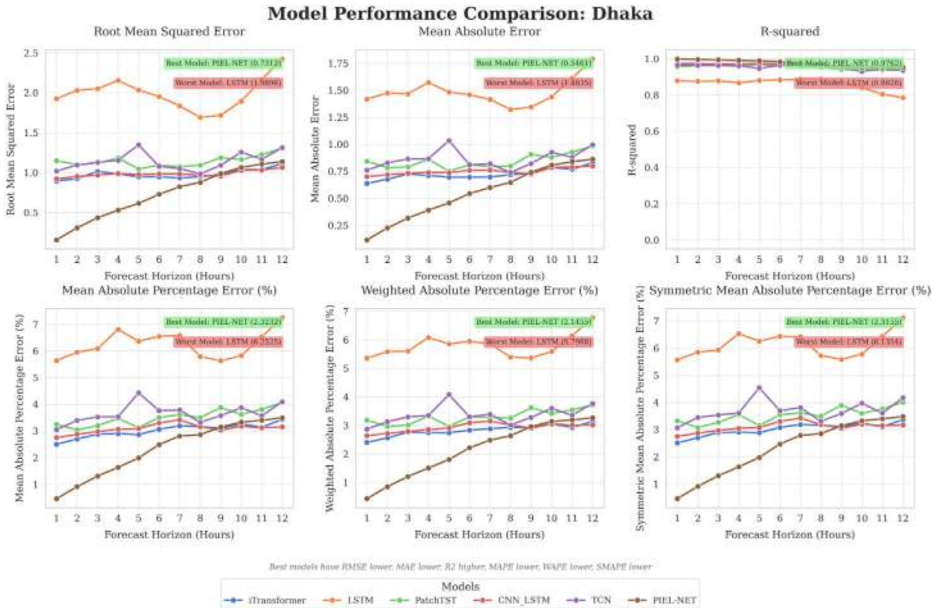

Fig. 12. Hourly comparison of error metrics across PIEL-NET and SOTA models for the Dhaka dataset.

4.5.4. Comparison of PIEL-NET with SOTA for the Tucumán dataset

Table 11 shows the performance of PIEL-NET and other SOTA models on the Tucumán dataset. As before, a lower RMSE indicates better performance in capturing extreme temperature events and PIEL-NET achieves the lowest RMSE (1.5524), outperforming models like iTransformer (2.1291) and TCN (2.0953). This highlights PIEL-NET's superior ability to handle temperature extremes. In terms of MAE, PIEL-NET again shows the best performance with an MAE of 1.0046, which is significantly lower than that of iTransformer (1.6801) and TCN (1.5975), confirming its consistent accuracy across all conditions.

When compared to other models, PIEL-NET also achieves the highest R-squared Score (0.9575), demonstrating its ability to explain the most variance in the data, followed by TCN (0.9256) and iTransformer (0.9221). The MAPE for PIEL-NET is 5.15% that is the lowest, unlike other models such as PatchTST (14.70%) and LSTM (9.52%), which show considerably higher relative errors, indicating less accurate performance on this metric. For WAPE, PIEL-NET's value of 4.37% is the lowest, underscoring its efficient

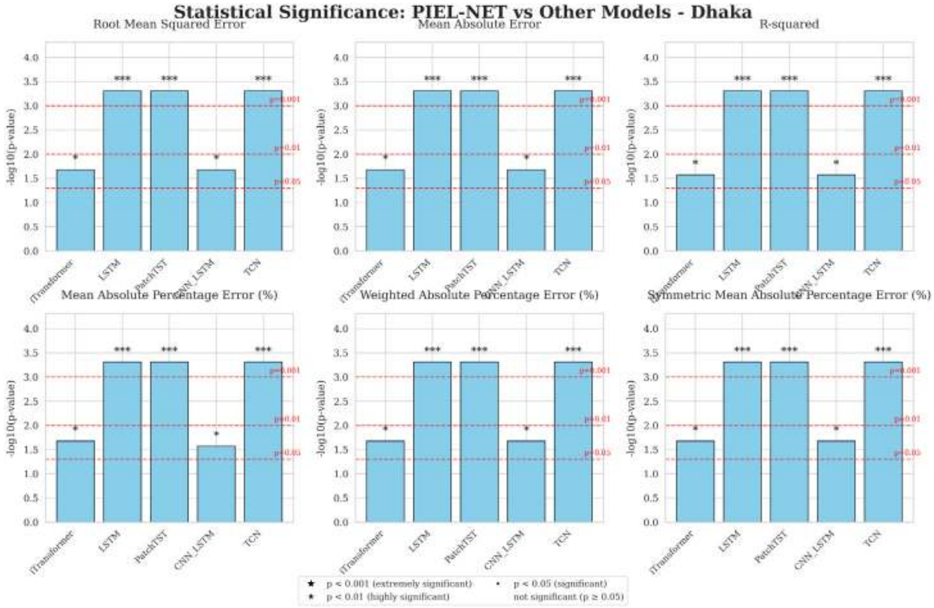

Fig. 13. Wilcoxon signed-rank test results confirming that PIEL-NET is significantly different from all other models.

Table 11
Performance comparison with SOTA models on the Tucumán dataset.

Model	RMSE (°C)	MAE (°C)	R-squared score	MAPE (%)	WAPE (%)	SMAPE (%)
iTransformer	2.1291	1.6801	0.9221	10.01%	7.33%	8.85%
LSTM	2.4874	1.9456	0.8988	9.52%	8.47%	9.08%
PatchTST	2.7914	2.2449	0.8724	14.70%	9.82%	12.13%
CNN_LSTM	2.4286	1.9047	0.9040	10.18%	8.30%	9.38%
TCN	2.0953	1.5975	0.9256	8.42%	6.97%	8.10%
PIEL-NET	1.5524	1.0046	0.9575	5.15%	4.37%	5.04%

handling of temperature extremes compared to iTransformer (7.33%) and LSTM (8.47%). In terms of SMAPE, PIEL-NET again leads with a value of 5.04%, outperforming iTransformer (8.85%) and TCN (8.10%).

Among the closest competitors, TCN and iTransformer show comparable results across most metrics, but still fall short of PIEL-NET, especially in capturing extreme temperatures and minimizing errors. While TCN demonstrates the second-lowest RMSE (2.0953), it lags behind PIEL-NET in terms of MAE (1.5975) and SMAPE (8.10%). iTransformer, while competitive with TCN in terms of RMSE and R-squared Score, still shows higher relative errors in MAPE (10.01%) and WAPE (7.33%).

The hourly comparison, shown in Fig. 14, illustrates PIEL-NET's robust performance across all periods, particularly during temperature extremes. Finally, the Wilcoxon signed-rank test results in Fig. 15 confirm that PIEL-NET's performance is statistically significantly different from that of the other models across all metrics, ensuring that the observed improvements are not due to random variations.

4.6. Comparison against closest competing model in extreme temperatures

The issue with traditional data models is that they rely heavily on the availability of substantial amounts of examples to perform well. However, extreme temperature events are rare and these infrequent occurrences often lead to poor model performance in such scenarios. The specialized agent for hard examples in our proposed PIEL-NET model addresses this limitation, allowing it to perform better during these rare yet crucial extreme temperature events. This section shows how PIEL-NET outperforms its closest competitor in capturing the nuances of extreme weather conditions, particularly in rare temperature events that are often neglected by standard models.

Fig. 16 shows the performance of the proposed model against its closest competitor(TCN) on the Beijing data. In this case, the temperature distribution subplot (top left of Fig. 16) shows drastically fewer samples below $-4.1\,°C$ and a similar scarcity above $27.4\,°C$, which defines sparse regimes. The MSE versus normalized sample density subplot (top right of Fig. 16) indicates that in sparse regions the closest competitor TCN reaches about $6\,°C^2$ while PIEL–NET is about $2.0\,°C^2$. The critical zone subplot (bottom left of Fig. 16) yields for $T < -4.1\,°C$ an MSE of $0.92\,°C^2$ for PIEL–NET and $1.76\,°C^2$ for TCN, which is a 47.3% reduction, and for $T > 27.4\,°C$ an MSE of $1.12\,°C^2$ for PIEL–NET and $2.34\,°C^2$ for TCN, which is a 52.14% reduction. The error distribution subplot visualizes these differences across the full sample range in $°C^2$.

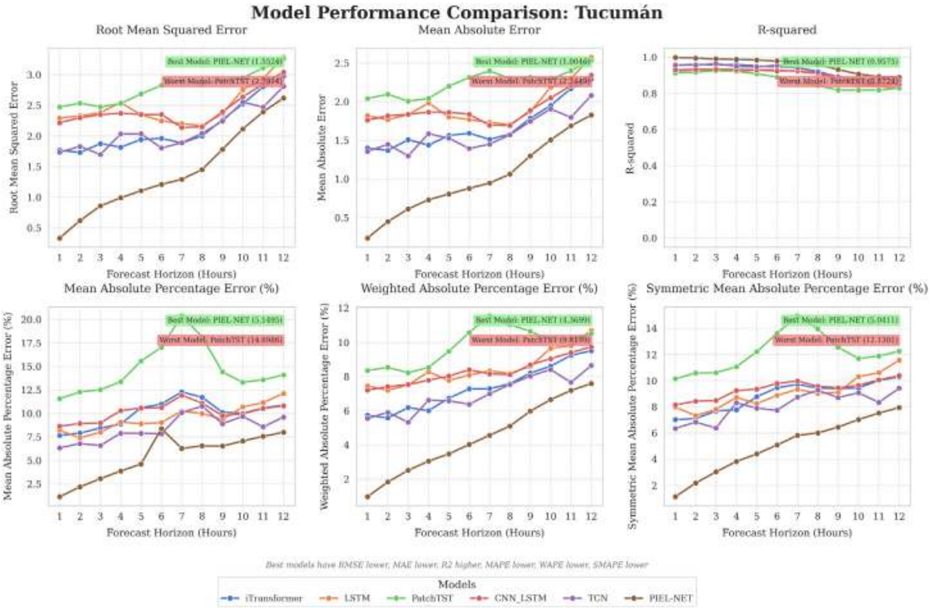

Fig. 14. Hourly comparison of error metrics across PIEL-NET and SOTA models for the Tucumán dataset.

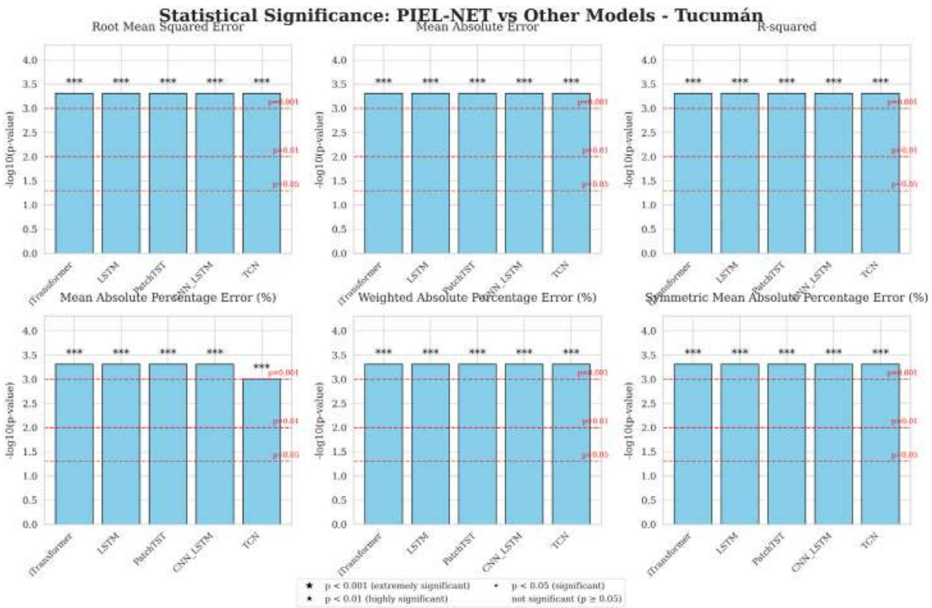

Fig. 15. Wilcoxon signed-rank test results confirming that PIEL-NET is significantly different from all other models.

Similarly, Fig. 17 compares PIEL–NET and its closest competitor, LSTM, on Calgary datasets. The temperature distribution subplot (top left, Fig. 17) shows drastically fewer samples below 4 °C, decreasing to about a few samples below −15.1 °C. Such low temperatures raise health risk and increase energy demand, so accuracy in these regimes is operationally important. The MSE versus normalized sample density subplot (top right, Fig. 17) shows that in sparse regions LSTM reaches about 30 °C^2, while PIEL–NET is about 3.0 °C^2. The critical zone subplot (bottom left, Fig. 17) yields for $T < -7.4$ °C an MSE of 2.39 °C^2 for PIEL–NET and 7.76 °C^2 for LSTM, a 69.2% reduction and for $T > 18.3$ °C an MSE of 1.37 °C^2 for PIEL–NET and 3.73 °C^2 for LSTM, a 63.27% reduction. The error distribution subplot (bottom right, Fig. 17) visualizes these differences across the full sample range in °C^2.

For the Dhaka dataset, Fig. 18 compares PIEL–NET and the closest competitor iTransformer. The temperature distribution subplot (top left, Fig. 18) shows drastically fewer samples below 20 °C and fewer samples per degree above 30 °C. Such high temperatures raise health risks and increase energy demand, so accuracy in these regimes is operationally important. The MSE versus normalized

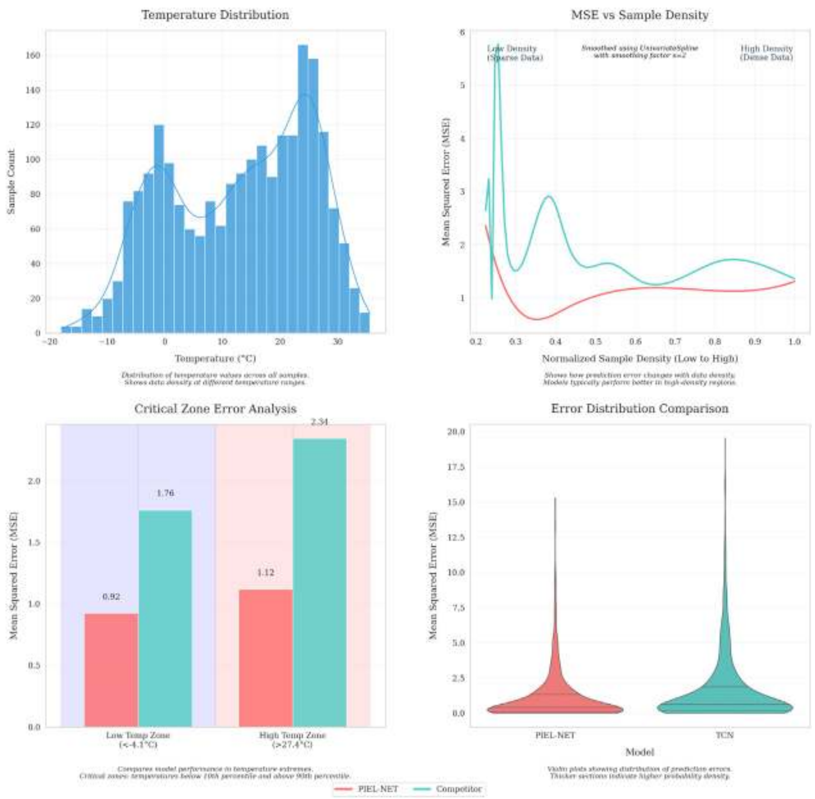

Fig. 16. Temperature distribution comparison for PIEL-NET and the best competitor on the Beijing dataset.

sample density subplot (top right, Fig. 18) shows that in sparse regions iTransformer reaches about 4.0 °C^2 while PIEL–NET is about 0.5 °C^2. The critical zone subplot (bottom left, Fig. 18) yields for $T < 18.3$ °C an MSE of 0.34 °C^2 for PIEL–NET and 0.69 °C^2 for iTransformer which is a 50.72% reduction and for $T > 30.2$ °C an MSE of 0.28 °C^2 for PIEL–NET and 1.76 °C^2 for iTransformer which is an 84.09% reduction. The error distribution subplot (bottom right, Fig. 18) visualizes these differences across the full sample range in °C^2.

Finally, for the Tucumán dataset, Fig. 19 compares PIEL–NET and the closest competitor, TCN. The temperature distribution subplot (top left, Fig. 19) shows almost a Gaussian behavior with very few samples under 12 °C and above 31 °C. In this case again, the high temperatures can raise health risks and increase energy demand, so accuracy in these regimes is operationally important. The MSE versus normalized sample density subplot (top right, Fig. 19) shows that in sparse regions TCN reaches about 4.0 °C^2 while PIEL–NET is about 0.6 °C^2. The critical zone subplot (bottom left, Fig. 19) yields for $T < 12.9$ °C an MSE of 0.60 °C^2 for PIEL–NET and 1.43 °C^2 for TCN, which is a 58.04% reduction, and for $T > 31.7$ °C an MSE of 0.99 °C^2 for PIEL–NET and 3.65 °C^2 for TCN, which is a 72.87% reduction. The error distribution subplot (bottom right, Fig. 19) visualizes these differences across the full sample range in °C^2.

Hence, the PIEL-NET model outperforms its closest competitors in handling extreme temperature events across multiple datasets. For instance, in the Beijing data, PIEL-NET achieves a 47.3% reduction in MSE for temperatures below −4.1, °C and a 52.14% reduction for temperatures above 27.4, °C. Similarly, in the Calgary dataset, it shows a 69.2% reduction for temperatures below −7.4, °C and a 63.27% reduction for temperatures above 18.3, °C. In the Dhaka dataset, PIEL-NET provides a 50.72% reduction for temperatures below 18.3, °C and an 84.09% reduction for temperatures above 30.2, °C. These results confirm PIEL-NET's improved performance in scenarios with rare temperature occurrences, showing its effectiveness in operational predictions systems.

4.7. Early warning system performance evaluation

To demonstrate the operational utility of PIEL-NET in urban climate risk management, this study evaluates its performance as the core component of an early warning system for temperature extremes. The evaluation assesses the model's capability to trigger timely alerts for impending cold snaps or heatwaves over a 12-h prediction horizon. The evaluation methodology first computes the average ground-truth temperature across subsequent 12-h windows for each temporal instance i:

$$A_i = \frac{1}{12} \sum_{h=1}^{12} T_i(h). \tag{42}$$

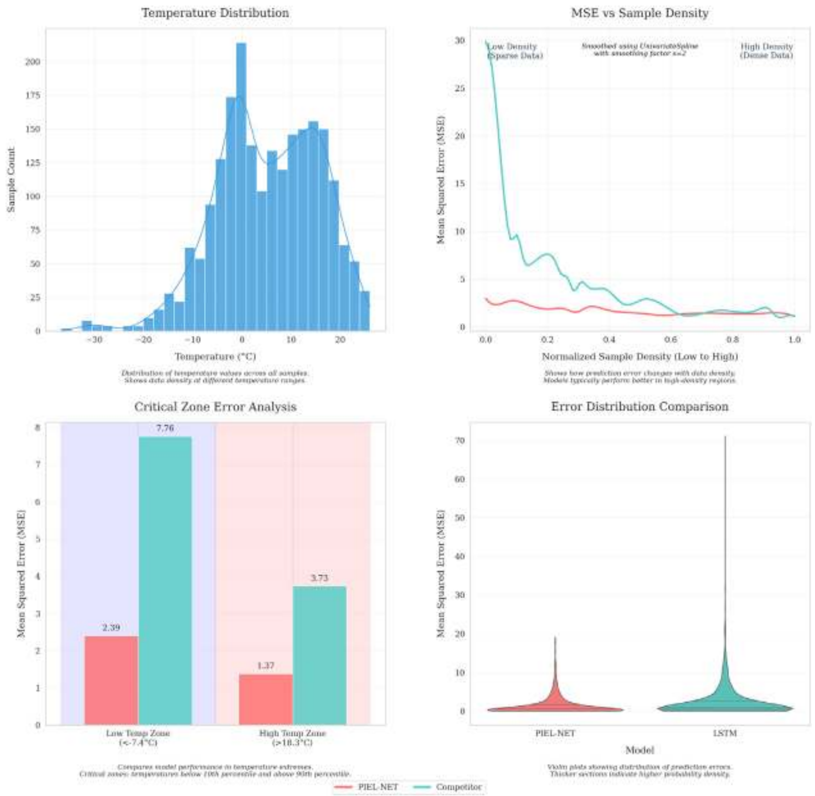

Fig. 17. MSE vs. sample density for the best models in the Calgary dataset.

where, $T_i(h)$ represents the observed temperature at lead hour h and A_i denotes the 12-h average temperature. Binary warning labels are assigned based on distributional extremes of the averaged temperatures:

$$y_i = \begin{cases} 1, & \text{if } A_i \le P_{10}(A) \text{ or } A_i \ge P_{90}(A) \\ 0, & \text{otherwise,} \end{cases} \tag{43}$$

where $P_q(A)$ indicates the qth percentile of the temperature averages $\{A_i\}$, with $y_i = 1$ flagging critical thermal conditions requiring alerts. These percentile thresholds define extremes relative to each city-center grid cell distribution in the reanalysis data and are operational markers for this study. Model predictions undergo an identical transformation to warning decisions:

$$\hat{A}_i^{(m)} = \frac{1}{12} \sum_{h=1}^{12} \hat{T}_i^{(m)}(h), \tag{44}$$

$$\hat{y}_i^{(m)} = \mathbf{1} \left\{ \hat{A}_i^{(m)} \le P_{10}(A) \text{ or } \hat{A}_i^{(m)} \ge P_{90}(A) \right\}, \tag{45}$$

where $\hat{T}_i^{(m)}(h)$ denotes the predicted temperature from model m at lead hour h, $\hat{A}_i^{(m)}$ represents the corresponding predicted average and $\hat{y}_i^{(m)}$ indicates the model's warning decision. Table 12 presents the early warning performance across all urban environments. PIEL-NET achieves the highest Dice coefficient and accuracy in every city, demonstrating superior reliability for operational deployment. In Beijing, PIEL-NET attains a 0.840 Dice score compared to 0.745 for the closest competitor (TCN), while maintaining balanced precision (0.834) and recall (0.845). The Calgary evaluation shows particularly strong performance with 0.899 Dice score and exceptional recall (0.891) for cold extremes, a critical requirement for preventing cold-related morbidity. Tropical conditions in Dhaka reveal PIEL-NET's robustness with 0.851 Dice score and optimal precision–recall balance (0.841 precision, 0.860 recall). Tucumán results demonstrate significant improvement in recall (0.727) over alternative models, reducing dangerous missed detections during rapid temperature transitions.

Notably, while iTransformer achieves high precision (0.968 in Beijing, 0.947 in Dhaka), its substantially lower recall (0.418 and 0.649 respectively) indicates critical under-detection of extreme events. This precision–recall imbalance proves operationally hazardous as missed extremes carry severe societal consequences. PIEL-NET's consistent superiority in recall-oriented metrics confirms its effectiveness in minimizing missed detections — the primary failure mode with highest impact in early warning applications. The balanced performance across diverse climate regimes demonstrates the framework's robustness for global urban deployment.

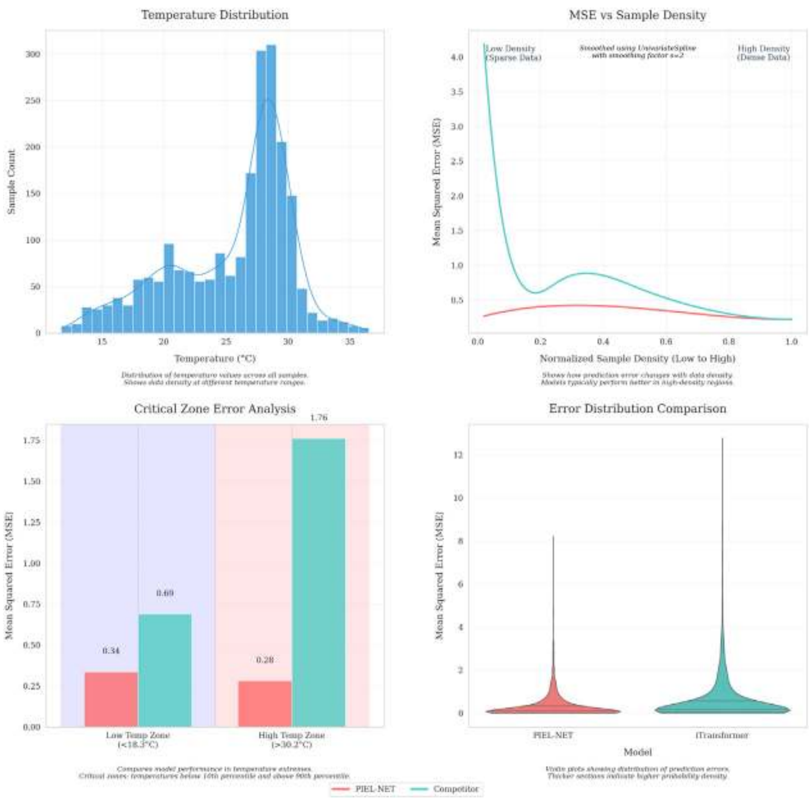

Fig. 18. Error analysis in critical temperature zones for the Dhaka dataset.

Table 12
Extreme event detection metrics for early warning systems. Best performance per city shown in bold.

City	Model	Precision	Recall	Accuracy	Dice
Beijing	PIEL-NET	0.834	**0.845**	**0.935**	**0.840**
	TCN	0.809	0.691	0.905	0.745
	LSTM	0.806	0.605	0.891	0.691
	CNN-LSTM	0.851	0.518	0.885	0.644
	PatchTST	0.762	0.568	0.877	0.651
	iTransformer	**0.968**	0.418	0.880	0.584
Calgary	PIEL-NET	0.907	**0.891**	**0.960**	**0.899**
	TCN	0.881	0.805	0.939	0.841
	LSTM	**0.910**	0.736	0.932	0.814
	CNN-LSTM	0.883	0.786	0.936	0.832
	PatchTST	0.677	0.827	0.886	0.744
	iTransformer	0.847	0.805	0.931	0.825
Dhaka	PIEL-NET	0.841	**0.860**	**0.939**	**0.851**
	TCN	0.799	0.824	0.922	0.812
	LSTM	0.694	0.459	0.849	0.553
	CNN-LSTM	0.917	0.748	0.935	0.824
	PatchTST	0.814	0.770	0.918	0.792
	iTransformer	**0.947**	0.649	0.921	0.770
Tucumán	PIEL-NET	0.874	**0.727**	**0.924**	**0.794**
	TCN	0.816	0.605	0.893	0.695
	LSTM	0.788	0.405	0.858	0.535
	CNN-LSTM	0.815	0.400	0.861	0.537
	PatchTST	0.743	0.500	0.864	0.598
	iTransformer	**0.926**	0.509	0.893	0.657

4.7.1. Early warning case study of extreme cold in Calgary

We present two twelve-hour windows in Calgary to show how the models behave as night turns into day. On the eighth of February in 2024, the air warms after sunrise, and PIEL NET follows that rise closely. On the twenty sixth of February in 2024,

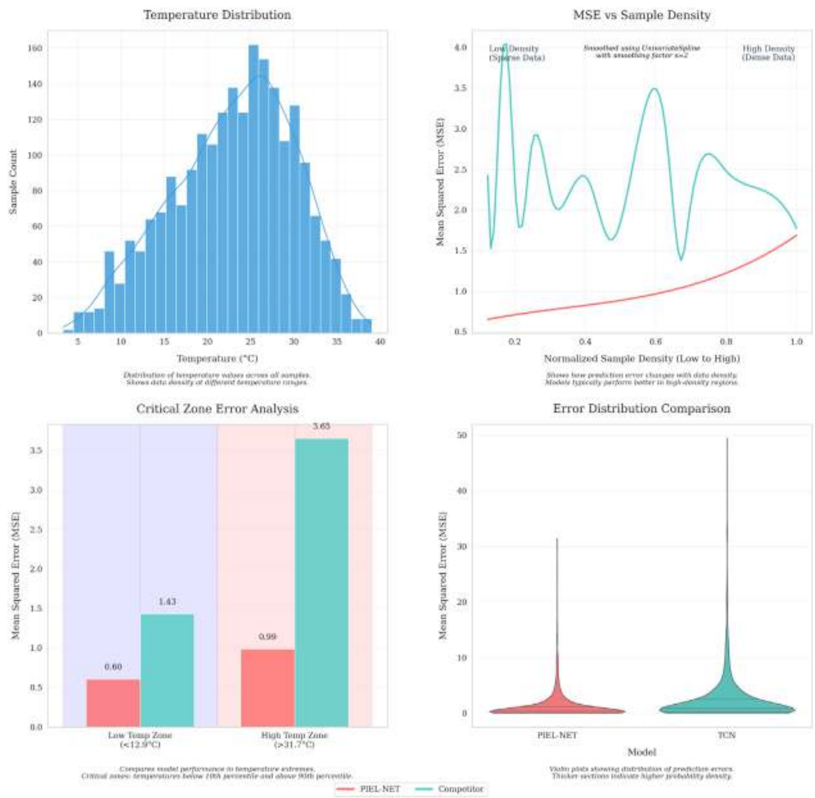

Fig. 19. Error distribution comparison for Tucumán dataset using violin plots.

Table 13
Calgary case study windows for twelve hours. Lower RMSE is better. Both windows triggered an early warning with **PIEL NET** while most competitors missed.

Event window	Twelve hour mean (°C)	Exceedance below P10 (°C)	PIEL NET RMSE (°C)	Next best (RMSE)
2024-02-08 00 h to 11 h	−9.05	1.66	**0.55**	iTransformer (2.29)
2024-02-26 00 h to 11 h	−12.93	5.54	**2.80**	LSTM (6.39)

the next day stays deeply cold with little rebound and PIEL NET anticipates this persistence, while other models follow the general trends that with a start of the day temperature will rise. Both windows are extreme since their twelve hour means sit below the Calgary P10 threshold which appears as a horizontal line in the figures. Each window contains seven night samples and five day samples so the night to day evolution is directly comparable (see Fig. 20, 21 and Table 13).

On the eighth of February in 2024 the morning warms steadily and PIEL NET stays closest hour by hour with absolute error near 0.53 at 09 h and near 0.48 at 11 h. Four of five competitors do not trigger the early warning. PatchTST is the only competitor that raises a warning and it paints a harsher picture by remaining far too cold during daylight which inflates hourly errors. On the contrary on the twenty sixth of February in 2024 the day shows little recovery after sunrise and PIEL NET keeps a cold outlook and limits warm bias while all five competitors warm strongly and miss the warning. The competing models appear to have learned a general pattern that daytime brings warming while the proposed model uses its meteorological inputs to recognize that this event would stay cold into the day just like the night and therefore generates the early warning ahead of time. These outcomes demonstrate that tail-focused learning combined with meteorological conditioning enables PIEL-NET to suppress false daytime rebound during persistent cold while faithfully tracking genuine post-sunrise warming when present. The model improves early warning recall and limits warm bias around sunrise, thereby reducing missed alerts and large hourly errors in the most critical hours.

5. Discussion

PIEL–NET improves the accuracy of the prediction in four geographically different cities in both mean and extreme conditions. The hierarchical design creates complementary strengths that the final fusion exploits. The physics terrain agent captures signals tied to advection and topography, which carry more weight in continental settings with large spatial gradients such as Calgary and Beijing. The spatio–temporal agent learns mesoscale patterns and smooth seasonal structure, which dominate in Dhaka and

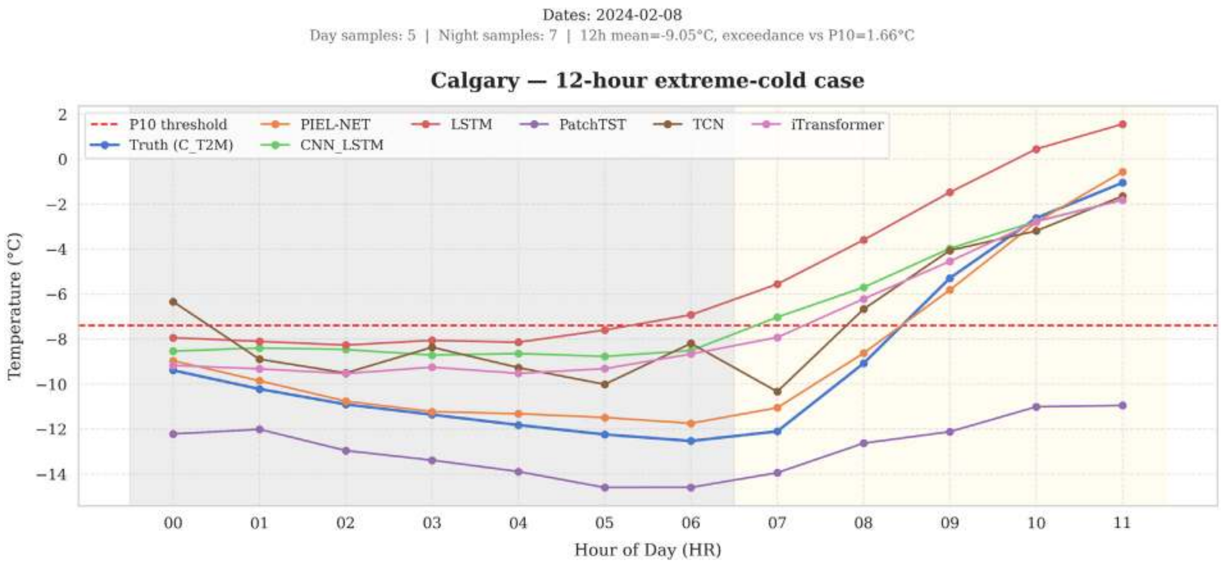

Fig. 20. Calgary twelve hour window on the eighth of February in 2024. PIEL NET tracks the post sunrise warming and stays closest to the truth. The P10 line marks the warning threshold. Night 00 h to 06 h and day 07 h to 11 h give 7 and 5 samples.

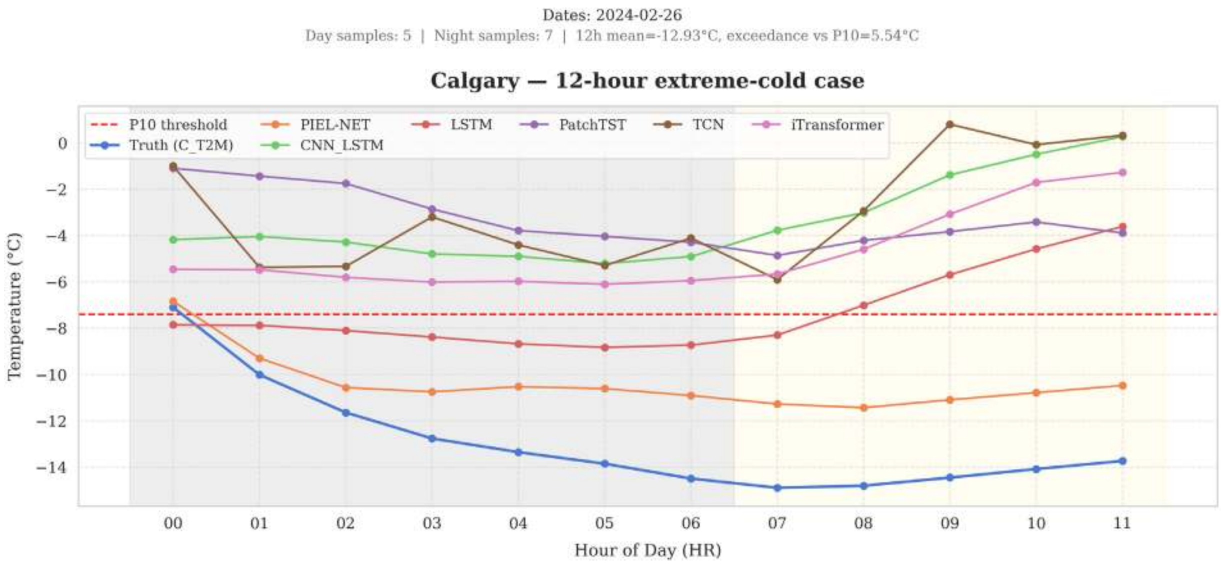

Fig. 21. Calgary, 12-h window on 26 February 2024. The day remains in deep freeze despite sunrise. PIEL-NET anticipates limited warming while other models warm too much. The P10 line marks the warning threshold. Night 00:00 to 06:00 and day 07:00 to 11:00 provide 7 and 5 samples.

are important in Tucumán. The regime–specialized agent, trained with Regime Adaptive Focal Loss, concentrates learning on the hard residuals that the other agents miss, which strengthens performance during cold snaps and heatwaves. The final fusion then aggregates these three views and delivers the lowest MAE in every city with gains over the best single agent of about 6.9% in Beijing, 3.4% in Calgary, 7.3% in Dhaka, and 4.2% in Tucumán.

Cross city behavior clarifies why components matter differently. In Beijing and Calgary, the terrain aware agent alone reaches MAE near 1.20 and 1.40 °C, close to the spatio–temporal baseline near 1.19 and 1.29 °C, and far better than the physics only model, which confirms the value of advection context even before fusion. In Dhaka, the spatio–temporal baseline is the strongest single agent with MAE 0.589 °Cwhile the terrain aware agent is a close second with MAE 0.601 °C, which is consistent with weak horizontal gradients and a dominant seasonal cycle. In Tucumán, which shows stronger transitions across seasons, the fused model reduces MAE from 1.0485 to 1.0046 °Cand also improves tail errors, which indicates that mixture learning helps when regimes shift over short periods. Tail improvements are larger than average gains and they align with the design of RAFL. In Calgary the model reduces mean squared error by 69.2% for temperatures below −7.4 °C. In Beijing the model cuts error by about 52% for temperatures above 27.4 °C and by about 47% for temperatures below −4.1 °C. In Dhaka the model reaches up to 84% reduction for temperatures above 30.2 °C. These gains track the intention to focus capacity on rare and operationally important regimes where data are sparse and physical priors alone underperform.

Early warning performance follows the same pattern. We define extremes using city-specific tenth and ninetieth percentiles of the twelve-hour mean temperature, then evaluate binary alerts over the twelve-hour horizon. PIEL–NET attains a Dice score of 0.840 in Beijing with precision 0.834 and recall 0.845, 0.899 in Calgary with precision 0.907 and recall 0.891, 0.851 in Dhaka with precision 0.841 and recall 0.860, and 0.794 in Tucumán with precision 0.874 and recall 0.727. High recall in Calgary and Beijing limits missed cold or hot events, which supports public health planning and grid reliability. Dhaka achieves a balanced tradeoff that supports actionable alerts without alarm fatigue. Tucumán shows strong precision and moderate recall, which reduces false alarms but leaves room to capture more fast transitions. These patterns suggest a practical path to stronger generalization. The mixture weights should be adjusted to flow and terrain indicators such as wind speed, wind direction persistence, local elevation gradients, and recent stability. Policy and operations can leverage the gains directly. City operators can schedule demand response, pre cool public buildings and shelters, and target outreach to vulnerable groups with better lead time. Energy planners can use the alerts to manage peak risk during heat or cold stress. Urban climate services can port the same framework to related variables such as humidity and apparent temperature, which drive heat health risk.

However, there are some limitations for this work. The study uses NASA POWER reanalysis at 0.5° × 0.625° with hourly cadence and a nine–point sampling layout with 100 km offsets around each city center. This configuration smooths urban heterogeneity, weakens street–canyon signals and land–water contrasts, and underrepresents local advection and topographic channeling. Station–level metadata and quality control flags do not enter the benchmarks, which supports cross–city comparability yet constrains fine–scale applicability. To test applicability future work can deploy a dense in situ network across neighborhoods and the peri–urban ring and then evaluate PIEL–NET under the same protocol in a city scale setting.

6. Conclusion

In operational terms two outcomes stand out. Extreme-temperature recall rises while mean accuracy holds across four climates, with tail MSE reductions of 69.2% for $T < -7.4\,°C$ in Calgary and up to 84% for $T > 30.2\,°C$ in Dhaka, and Dice reaching 0.899 over a 12-h horizon. The same Beijing-tuned configuration transfers across cities without per-site retuning, which supports scalable use. The compute budget remains small at 1.611 GFLOPs and 2.222 M parameters, and we verify edge feasibility on a Jetson Nano dev kit. Together these results indicate that regime-focused learning with a light physical prior improves early-warning reliability at the city-center grid cell while remaining practical for resource-constrained settings.

This study was carried out entirely in a simulation environment. We used gridded reanalysis data from NASA's POWER archive (Stackhouse et al., 2015), which are available only at a coarse 0.5° × 0.625° resolution. The sampling points, spaced 100 km apart from each city center, were selected such that each peripheral point aligned with a distinct grid cell. Future research will include the deployment of a physical sensor network at intervals of less than 100 km to validate these results under real-world conditions. This validation will involve installing meteorological stations at both urban centers and peripheral locations to capture fine-scale temperature variations and urban heat island effects.

Looking ahead, our future work will focus on three key directions. First, we will develop real-time deployment strategies for operational forecasting systems, optimizing computational efficiency for rapid prediction updates. Second, we will integrate PIEL-NET with urban planning and decision support tools, creating actionable interfaces for heat-health warning systems and energy demand management. Third, we will extend the framework to incorporate additional climate variables including humidity, wind speed, and precipitation, enabling comprehensive multi-variable urban environmental prediction. These advancements will enhance the model's practical utility for climate resilience planning and sustainable urban development.

CRediT authorship contribution statement

Laeeq Aslam: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Runmin Zou:** Supervision, Resources, Project administration. **Gang Li:** Writing – original draft, Methodology, Conceptualization. **Ebrahim Shahzad Awan:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Sara Mouafik:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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All authors reviewed and approved the final manuscript.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.uclim.2025.102669.

Data availability

The data used in this study is publicly available from NASA's POWER project (https://power.larc.nasa.gov/).

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